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Strategic transformation of the sales function in servitization: A multi-level configurational view

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ABSTRACT

This study explores how manufacturing firms reshape their sales function in their shift to a service business (*servitization*). Drawing on polar-type case studies at different stages of servitization, this research analyses how the sales of base and advanced services differ in terms of organisational structures, roles and responsibilities of sales teams, processes and tools, and knowledge and skills. The research reveals that standardised approaches and centralised governance often constrain the evolution of advanced services sales. We argue that modularity can also hinder the strategic development of sales functions in early-stage servitization. In contrast, it drives legitimation of professional sales teams in more mature contexts. Moreover, digital tools that support the selling of base and advanced services can differ significantly, with the latter benefiting from a more sophisticated portfolio of information systems and technologies such as simulators and configurators, while the former remain anchored to legacy tools supporting the sales of product business. The paper contributes to the servitization and industrial marketing literature by offering a multilevel view of the transformation of sales functions in connection with servitization, highlighting the interplay between standardisation, autonomy, modularity, and digitisation as impediments or enablers for the sale of advanced services.

1. Introduction

Manufacturing firms are increasingly augmenting their product-centric offerings with different kinds of services (Baines et al., 2020), which require greater customer collaboration and long-term engagement (Ulaga & Reinartz, 2011). This strategic transformation—entailing significant changes to traditional business models—is commonly termed 'servitization' (Baines et al., 2024). A key distinction in this domain lies between the offering of base rather than advanced services (Baines et al., 2020). Base services (e.g. installation, spare parts provision, repair) are reactive and support product functioning and restoration. Conversely, advanced services such as full-risk maintenance contracts —namely service agreements in which the provider assumes primary responsibility for ensuring agreed outcomes or performance levels over time, along with the associated operational and financial risks — guarantee certain outcomes or performances which are critical for the customer (Raddats et al., 2019; Rapaccini, 2015). This evolution affects the firm's value proposition and requires profound organisational and cultural transformation (Spadafora & Rapaccini, 2024). The shift from base to advanced services requires manufacturers to adopt new operational logics (Tuli et al., 2007), often involving increased customisation, digitalisation, and long-term commitments (Favoretto et al., 2022; Khanra et al., 2021). Recent research demonstrates the strategic relevance of servitization in worldwide manufacturing sectors (Mastrogiacomo et al., 2019) and the growing revenues from advanced services (Arioli et al., 2025). This is in line with the growth of the literature dealing with servitization, which reflects the increasing interest of the academic community in this phenomenon (Khanra et al., 2021).

The growing scholarly interest in servitization has largely been catalysed by advances in digital technologies, which have accelerated the growth of service offerings (Raddats et al., 2019) and given rise to the notion of digital servitization (Paschou et al., 2020) —defined as the intertwining of digitalisation and servitization to enable, enhance, and scale service-based value creation (Favoretto et al., 2022; Khanra et al., 2021; Kohtamäki et al., 2020; Tronvoll et al., 2020). Digital technologies—such as IoT connectivity and data analytics—act as critical enablers by generating continuous usage data, supporting remote monitoring and predictive maintenance, and underpinning outcome-based contracts and risk-sharing arrangements (Cenamor et al., 2017;

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Rapaccini et al., 2020; Sklyar et al., 2019). These mechanisms do not merely expand the service portfolio. They reshape commercial logics at the customer interface (e.g., value quantification, performance assurance, and multi-actor coordination), thereby compelling a substantive reconfiguration of the sales function (Sklyar et al., 2025).

In this context, the sales function represents a critical yet often problematic aspect (Kowalkowski et al., 2025). Unlike traditional product sales, the sale of advanced services requires engaging customers in longer, more consultative interactions, articulating value-in-use rather than features, and co-developing solutions that often require coordination across departments (Kaski et al., 2018; Kindström et al., 2015; Ulaga & Loveland, 2014). Sales teams are expected to develop new capabilities while operating within restructured organisational processes and systems (Sklyar et al., 2025; Terho et al., 2012). Research has highlighted the need for new tools such as CRM systems, calculators of customer value and configurators (Rapaccini, 2015). These latter are crucial for complex solution selling, as they facilitate the acquisition of customer requirements that are then input to the design of feasible product-service solutions (Panagopoulos et al., 2017; Rapaccini et al., 2020). Yet the challenge extends beyond individual adaptation: at the organisational level, firms must reconcile conflicting service models and sales logics (Kowalkowski et al., 2025). Base services are often standardised and transactional, while advanced services require tailored offerings, strong cross-functional collaboration, and a richer and more complex set of digital technologies and tools (Favoretto et al., 2022; Sklyar et al., 2025). This coexistence creates internal tensions—incentive misalignment, resource allocation dilemmas and cultural clashes—that sales organisations are often underprepared to address (Brax et al., 2017; Kindström & Kowalkowski, 2014). For instance, sales units may struggle to balance productivity targets tied to base services with the time- and knowledge-intensive demands of selling more advanced services.

Despite these insights, important theoretical gaps remain. First, although prior research has examined sales function in servitization at different levels of analysis—such as organisational structures, and individual sales skills—these perspectives have largely developed in parallel (Kindström et al., 2015; Kowalkowski et al., 2025; Rabetino et al., 2018; Sklyar et al., 2025). As a result, existing theory offers a limited understanding of how changes at the strategic and organisational levels interact with, and shape, the transformation of the sales function during the shift from base to advanced services (Favoretto et al., 2022; Kohtamäki et al., 2025). This fragmentation hampers the development of a multilevel theorisation of servitization, leaving insufficient understanding of how alignment of practices and tools across organisational structures unfold over time (Salonen et al., 2021; Tuli et al., 2007).

Second, although scholars increasingly acknowledge that manufacturers rarely complete a linear transition toward advanced services and instead operate hybrid service portfolios in which base and advanced services coexist (Kowalkowski et al., 2015, Kohtamäki et al., 2025; Storbacka et al., 2013; Windahl & Lakemond, 2010), the implications of such hybridity for the organisation and governance of the sales function remain insufficiently theorised (Kowalkowski et al., 2025; Sklyar et al., 2025). Existing studies recognise the presence of competing service models and sales logics (Ulaga & Loveland, 2014; Ulaga & Reinartz, 2011), yet provide limited explanation of how sales organisations structurally accommodate, coordinate, and legitimise the coexistence of base and advanced services (Brax et al., 2017; Kindström & Kowalkowski, 2014). As a result, current theory offers only partial insight into the organisational mechanisms through which sales functions manage persistent tensions arising from hybrid service portfolios, constraining a more nuanced understanding of ambidexterity in servitized manufacturing contexts (Salonen et al., 2021; Visnjic et al., 2016).

Third, within the expanding literature on digital servitization, digital technologies are predominantly conceptualised as broad enablers of service growth, scalability, and performance (Kohtamäki et al., 2020; Tronvoll et al., 2020). While prior research highlights the importance of digital infrastructures such as IoT connectivity, data analytics and digital platforms (Ardolino et al., 2018), comparatively less attention has been devoted to theorising how specific digital tools and systems reconfigure commercial practices in distinct ways across base and advanced services, and how these differences shape the organisation and governance of the sales function (Favoretto et al., 2022; Rapaccini et al., 2020). This limits the understanding of the role of digital technologies in mediating complexity, supporting coordination, and managing internal tensions within sales organisations operating under conditions of service hybridity (Kohtamäki et al., 2025; Sklyar et al., 2025).

In addition to the theoretical gaps identified above, a recent systematic literature review by Kowalkowski et al. (2025) shows that research on selling and sales management in servitization is still largely conceptual, with limited empirical insight into how sales functions evolve in practice.

By addressing these underexplored areas, the current study aims to investigate how the sales function evolves to support the transition from base to advanced services, with particular attention to the organisational changes and internal tensions that emerge during this shift. In particular, the following research questions have guided our research: a) How does the transition from base services to advanced services impact the organisational structures, the processes and tools of the sales function, and the knowledge and skills of the salespersons in servitized manufacturing firms? and b) What tensions emerge within the sales function when base and advanced service offerings coexist?

To address the mentioned questions, we adopt a comparative case study approach, examining two manufacturing firms at different stages of servitization. Servitization stages are evaluated according to the Baines staircase (Baines et al., 2024), which distinguishes among base, intermediate, and advanced services. Base and intermediate services are provided with a product-dominated mindset, that is, helping customers in achieving resources and capabilities for product recovery and optimal operations. With the advanced services, conversely, the service provider takes the risk to obtain business outcomes, such as productivity or availability, and the customer pays in proportion to the results achieved.

In doing this, the paper makes three contributions. First, it theorises a sales transformation in servitizing manufacturers as a multilevel alignment process across organisational structures, processes and tools, and knowledge and skills. Second, it specifies the hybrid configurations and mechanisms that make concurrent operations of transactional (base) and consultative (advanced) logics workable within the same sales system. Therefore, it provides details about coordinated choices in mandates and governance mechanisms of the sales function. Third, it advances a systemic view of sales function development, showing that consultative behaviours consolidate when organisational legitimacy and appropriate autonomy are matched with the adoption of dedicated tools that differentially support base and advanced service selling across the customer journey.

The remainder of the article is structured as follows. Section 2 reviews prior research on changes to the sales function in servitizing firms. Section 3 describes the research methodology, including case selection, data collection, and procedures for data analysis and interpretation. Section 4 presents the empirical findings from the two case studies, focusing on each case study's changes in the organisational structures, processes and tools, knowledge and skills of sales functions. Section 5 discusses the key insights emerging from the comparison of the polar cases. Building on this, section 6 develops a novel multilevel framework for sales transformation in servitizing firms. Finally, Section 7 concludes the paper by highlighting its theoretical and managerial contributions, outlining its limitations, and suggesting avenues for future research.

2. Theoretical background

2.1. Sales transformation in servitization

The evolution of manufacturing firms toward servitization is widely

recognised as a staged and non-linear process that unfolds along a continuum of increasingly sophisticated service offerings (Baines et al., 2024; Raddats et al., 2019). Rather than a binary shift from products to services, servitization involves the progressive reconfiguration of firms' value propositions through base, intermediate, and advanced services (Oliva & Kallenberg, 2003; Rabetino et al., 2018). This typology has become foundational in the servitization literature, being characterised by increasing levels of relational intensity, risk sharing, and performance responsibility (Baines et al., 2024; Rapaccini & Visintin, 2015; Visnjic et al., 2016).

Base services—such as installation, spare parts provision and fix and repair—adhere to a product-centric logic and are typically transactional, standardised, and reactive (Kindström & Kowalkowski, 2009; Salonen & Jaakkola, 2015; Ulaga & Reinartz, 2011; Zhang & Banerji, 2017). Intermediate services include life-cycle support such as diagnostics, service agreements, and remote monitoring, representing more proactive yet still codified engagements (Favoretto et al., 2022; Paiola et al., 2024). Advanced services, by contrast, are outcome-based and involve contractual commitments on performance indicators such as uptime, availability, productivity, or energy savings (Rapaccini et al., 2020). Offering advanced services requires deep integration with customers' operations and a value co-creation approach built on long-term relationships (Kowalkowski et al., 2025; Ulaga & Loveland, 2014).

This progressive shift toward the offering of more advanced services entails not only strategic repositioning but also deep cultural and organisational transformations (Cron & Decarlo, 2010; Oliva & Kallenberg, 2003). As manufacturing firms expand their roles beyond traditional product delivery—often assuming activities previously performed by downstream actors and customers (Capanni et al., 2025; Spring & Araujo, 2013)—they face the challenge of redefining internal boundaries, responsibilities, and engagement logics (Grönroos & Voima, 2013).

These changes are evident within the sales function, which must evolve from transactional selling to consultative, relational practices. Selling base services can typically be managed through linear, codified routines—such as responding to requests for proposals, issuing quotations, and negotiating contracts (Kindström et al., 2015; Sheth & Sharma, 2008). In contrast, advanced services require dialogic and co-creative approaches, in which salespeople work closely with technical teams and customers to explore needs, design solutions, and articulate complex value propositions based on value-in-use (Panagopoulos et al., 2017; Sharma, 2007; Terho et al., 2012; Ulaga & Loveland, 2014). Sales teams must become more customer-embedded, contributing to the development of personalised, idiosyncratic solutions (Raddats et al., 2019; Spadafora et al., 2024; Tuli et al., 2007) and integrating firm-specific and contextual knowledge into the selling process (Powers et al., 2016; Windler et al., 2017). This shift requires not only new practices and tools (Rapaccini, 2015), but also a profound change in sales mindset and culture (Kowalkowski & Ulaga, 2017; Piercy, 2010; Ulaga & Kohli, 2018). Yet in practice, base, intermediate and advanced services simultaneously coexist. Their interplay creates internal complexity and tension, demanding an ambidextrous sales function that balances standardisation with customisation, productivity with responsiveness, and control with autonomy (Kindström & Kowalkowski, 2014; Kowalkowski et al., 2025; Rabetino et al., 2018).

To interpret how the sales system adapts to this layered and hybrid transformation, this research adopts the multilevel theoretical framework of Kozlowski and Klein (2000). This framework conceptualises organisations as hierarchical systems composed of interdependent levels, namely micro, *meso* and macro levels. Building from the presented literature review on this topic, we identify the following levels of transformation in the sales function, which we interpret as the loci of the corresponding multilevel adaptation:

1) Organisational structures—which reflect macro-level transformations such as changes to roles, responsibilities and configurations of the sales organisation;
2) Processes and tools—which reflect meso-level adjustments to routines as well as the adoption of new tools and practices;
3) Knowledge and skills—which reflect micro-level changes in individual knowledge and competencies.

The interaction among these domains is not linear or sequential. Rather, changes at one level can trigger, enable, or constrain changes at other levels. For instance, the adoption of a new CRM system (meso level) may require structural reorganisation (macro level) as well as new training of salespersons (micro level). In this sense, we conceptualise the evolution of the sales function as a multilevel reconfiguration that gives rise to new, emergent configurations. These represent an integrated outcome of simultaneous changes across multiple levels: structural and role-related transformations at the macro level, new processes and tools at the meso level, and evolving skillsets at the micro level. Together, these interdependent shifts shape a systemic pattern of organisational, technological, and individual adaptation that enables the sales function to respond effectively to the demands of servitization.

The following sections first elaborate on each domain (macro, meso, micro) and then synthesise the extant literature and provide a positioning of this paper.

2.2. Macro level: changes to organisational structure

Servitization requires a redefinition of how the sales function is structured and governed (Gremyr et al., 2010; Raddats & Burton, 2011). Some studies suggest the creation of a dedicated and independent service sales unit, distinct from product-oriented commercial structures (Gebauer & Kowalkowski, 2012; Khanra et al., 2021). This structural decoupling is justified by the fact that service sales require higher levels of customer orientation and embeddedness (Storbacka, 2011), as observed in professional service firms such as consultancies (Arnett & Badrinarayanan, 2005). To respond to the demand of complex service engagements, manufacturers increasingly introduce roles such as Global Account Teams, which manage cross-border accounts and ensure consistency in customer engagement (Kowalkowski et al., 2017; Sheth & Sharma, 2008).

Sales departments often exhibit resistance to the changes of servitization, as they are frequently affected by *siloed thinking*—that is, a rigid adherence to legacy product-centric structures, incentives, and routines that inhibit cross-functional collaboration (Kindström & Kowalkowski, 2014). Overcoming these silos requires a cultural and structural shift that fosters shared goals and cross-departmental coordination (Storbacka et al., 2009). Moreover, redesigning the incentive system becomes essential, as service sales typically generate value over longer horizons and through more complex interactions (Ulaga & Loveland, 2014). When compensation schemes remain aligned with short-term product turnover, salespeople are less inclined to invest in relational and consultative selling (Kindström & Kowalkowski, 2009; Reinartz & Ulaga, 2008). In this context, aligning incentives with strategic objectives is crucial for individual and organisational change (Kindström et al., 2015).

The sales functions of servitizing firms must also tackle the requirements of external actors, as the delivery of advanced services implies deeper and broader engagement with the customer organisation. Therefore, salespeople must mediate among multiple stakeholders with divergent expectations (Piercy & Lane, 2005; Selnes, 1998) while simultaneously acting as brand ambassadors who strengthen the manufacturer's reputation as a service provider (Sarasvuo et al., 2023). Trust-building becomes critical for cultivating long-term customer loyalty (Raja et al., 2013; Shams et al., 2024).

Beyond managing relationships, service-oriented sales teams are responsible for continuously sensing customer needs. This entails using

ethnographic methods such as interviews and focus groups to develop a nuanced understanding of client-specific operational, technological, and regulatory contexts (Kohli & Jaworski, 1990). Salespeople in servitizing firms must therefore act as problem solvers and consultants, rather than transactional agents (Grönroos & Voima, 2013; Storbacka et al., 2009; Tuli et al., 2007), using every customer interaction as an opportunity to surface latent or unmet needs (Sarasvuo et al., 2023).

At a strategic level, the sales function must also coordinate internal and external actors. Internally, this involves the mobilisation of colleagues across departments—including engineering, finance, and legal—to co-develop feasible and value-adding solutions (Kowalkowski et al., 2025; Panagopoulos et al., 2017). Externally, the sales team may need to integrate downstream partners such as distributors and dealers into the value delivery process (Capanni et al., 2025). Salespeople also play a role in dynamically managing the customer relationship portfolio, adapting strategies to anticipate shifts in buyer behaviour and expectations (Kindström & Kowalkowski, 2014; Teece et al., 1997). Ensuring stability in account assignments further supports the accumulation of customer-specific knowledge and enables the co-creation of tailored offerings (Storbacka, 2011; Tuli et al., 2007). Another critical change is the stricter integration between sales and engineering departments, to support modular product-service design. Modularity enables the configuration of customised solutions through standardised components, which helps balance flexibility and internal efficiency (Brady et al., 2005; Brax et al., 2017). In this regard, modular service architectures can act as theoretical and practical scaffolds that guide salespeople in aligning customer needs with feasible technical solutions (Favoretto et al., 2022; Rapaccini et al., 2023). However, if applied too rigidly, modularity may also constrain the autonomy of salespersons and reduce their ability to co-create value with customers (Kowalkowski et al., 2025; Sklyar et al., 2025). As such, the governance of modularity becomes key to enable consultative selling while preserving operational coherence.

In sum, servitization expands the role of the sales function beyond its traditional boundaries, requiring structural adaptation, cultural change, internal and external coordination. In line with our theoretical lens, these transformations correspond to macro-level changes in organisational structures and governance mechanisms (Kozlowski & Klein, 2000). When viewed through a multilevel perspective, these macro changes interact dynamically with meso- and micro-level adaptations, contributing to the emergence of new selling configurations.

2.3. Meso level: changes to processes and tools

Selling advanced services requires more structured processes than selling products and base services (Kohtamäki et al., 2020; Reinartz & Ulaga, 2008). Witell et al. (2020) propose 4 phases, respectively a) pre-bid engagement, b) negotiation, c) implementation, and d) delivery. In pre-bid engagement, sales teams activate the relationships necessary to identify the customer's needs, prior to their request. The negotiation phase focuses on customising the solution and defining the contractual terms. The implementation includes installation and starting-up the solution, while the delivery phase focuses on continuous support and performance monitoring (Brady et al., 2005; Tuli et al., 2007).

Extending the scope of the sales process implies greater coordination across customer interfaces and internal functions, underpinned by adaptive and iterative workflows (Bigdeli et al., 2017; Kowalkowski et al., 2017). Customer Relationship Management (CRM), Contract Management Systems (CMS) and Field Service Management (FSM) information systems can play a pivotal role in enabling this integration, thus increasing the visibility throughout the sales and delivery journey (Rapaccini et al., 2023; Sheth & Sharma, 2008). However, the full automation of these workflows remains limited. Key touchpoints of customer engagement still rely heavily on the relational and interpretive capacity of human actors (Kindström et al., 2013; Sklyar et al., 2025).

Advanced analytics support service-sales decisions by converting data collected from connected machines into insights, thus favouring the development of condition monitoring, diagnostic and predictive maintenance services (Ardolino et al., 2018). These capabilities are more and more delivered with the help of machine-learning and artificial intelligence tools, that enable dynamic pricing, contract design and field interventions optimisation (Sjödin et al., 2019). Moreover, data-driven tools enhance segmentation, customer profiling, demand forecasting, and contract optimisation (Allison & Flaherty, 2020; Kohtamäki et al., 2020), thus contributing to real-time responsiveness and personalised value propositions (Lenka et al., 2017). Simulation tools, augmented reality, and digital configurators can also support salespeople in illustrating, in a more tangible way, the value of advanced services (Rabetino et al., 2018; Rapaccini, 2015).

Nevertheless, digital tools are not sufficient on their own. A growing body of research shows that the real value of these technologies lies in their socio-technical integration—that is, how they are embedded in collaborative routines and knowledge-sharing processes (Coreynen et al., 2017; Kohtamäki et al., 2025). Sales professionals need to interpret insights, contextualise them, and act upon them in a coordinated manner. Therefore, technology adoption must be coupled with organisational learning and process redesign to achieve effective transformation (Sklyar et al., 2025; Zhang & Banerji, 2017).

This transformation also entails changes in governance and coordination mechanisms. In traditional selling, sales operated relatively autonomously. Conversely, in servitized contexts, the cross-functional coordination becomes essential. Processes must accommodate input from engineering, operations and finance, especially to support tailored pricing and deployment of product-service offerings (Kindström & Kowalkowski, 2009; Rapaccini, 2015). These collaborative requirements have led many firms to adopt hybrid governance structures, agile sales practices, and modular architectures that enable fast adaptation and co-innovation with clients (Lenka et al., 2017; Kowalkowski et al., 2017).

Despite this evolution, extant literature has historically placed more emphasis on individual skills and salespeople's capabilities (e.g., Ulaga & Loveland, 2014) than on the actual reconfiguration of sales practices (Kohtamäki et al., 2020; Kowalkowski et al., 2025; Rabetino et al., 2018; Sklyar et al., 2025). Recent studies have begun addressing this gap by focusing on how sales processes are practically transformed, including challenges in onboarding digital technologies, resisting inertia, and aligning KPIs with service-based logic (Sklyar et al., 2025). Still, further empirical research is needed to understand how organisations orchestrate such transitions across departments and over time (Khanra et al., 2021; Kowalkowski et al., 2025). In addition, service sales must accommodate a diversity of customer expectations, calling this for higher degrees of flexibility, modularity and organisational learning (Coreynen et al., 2017; Rabetino et al., 2018).

From a theoretical perspective, these developments correspond to the *meso* level of the multilevel framework guiding this study (Kozlowski & Klein, 2000). Specifically, they relate to the evolution of shared routines, norms and technologies, through which individual-level behaviours are translated into systemic organisational transformation. Understanding these intermediate mechanisms is therefore critical for conceptualising how selling capabilities evolve in response to servitization demands—not simply through new individual skills or structural reconfiguration, but through processual and technological adaptation embedded in day-to-day operations.

2.4. Micro level: changes to the knowledge and skills of salespeople

Salespeople of servitizing firms must learn new skills (Ulaga & Loveland, 2014). The development of more advanced capabilities—encompassing customer understanding, consultative selling, value-based communication, solution configuration, and cross-functional collaboration—is essential (Blocker et al., 2012; Rapaccini et al., 2023; Sklyar et al., 2025; Terho et al., 2012). Sellers must learn to

identify latent customer needs, co-develop tailored solutions and articulate the long-term benefits from partnering with the equipment producer. Reskilling of sales professionals is therefore crucial (Storbacka, 2011; Ulaga & Loveland, 2014).

The requested changes are impactful, and turnover among sellers is reported to be high (Reinartz & Ulaga, 2008). On average, less than a third of experienced salespeople is expected to embrace the new skillsets that selling advanced services imposes (Kowalkowski & Ulaga, 2017). The sellers of advanced services should master in-depth knowledge of customers' business models (Terho et al., 2012; Tuli et al., 2007), which is key to identifying the most suitable solutions to propose (Blocker et al., 2012) and to communicate effectively their value (Ulaga & Reinartz, 2011). In addition, they should develop abilities to manage more articulated relationships toward different interfaces of the client organisation, such as operations and IT departments (Rapaccini et al., 2023). As previously noted, customers are expected to actively contribute to the ideation of integrated solutions (Tuli et al., 2007); therefore, the sellers of advanced services must be able to translate even the most tacit and hard-to-codify customer needs into documented requirements. In this role, they must be able to recognise the boundary between what can and what cannot be accepted, refusing any requirement which would compromise, in the long run, both the provider's profitability and the customer's satisfaction (Raja et al., 2020; Paiola et al., 2024; Salonen et al., 2021).

Specific attitudes of salespersons, such as lifelong learning orientation, intrinsic motivation and emotional intelligence, can be a good ground for this change (Ulaga & Loveland, 2014). Recent contributions also underline the role of digital literacy and data-driven decision-making as critical skills for advanced service selling, especially in industrial contexts undergoing digital transformation (Kohtamäki et al., 2020; Sklyar et al., 2025).

Sellers of advanced services must interact with people delivering and receiving services (Eggert et al., 2020; Kowalkowski et al., 2025; Payne et al., 2008), as they are expected to collect feedback that can improve the quality of service delivery (Grönroos & Voima, 2013; Tronvoll et al., 2020). All this requires frequent and in-depth interactions between the sales teams and the field organisations of customers and downstream actors (Capanni et al., 2025; Kindström & Kowalkowski, 2009; Koponen et al., 2019; Raja et al., 2020; Sjödin et al., 2019). In this action, sellers are also called upon to overcome the linguistic and cultural barriers that are typical of global businesses (Parida & Jovanovic, 2022; Storbacka et al., 2009; Tippmann et al., 2023). Finally, they need strong leadership for project coordination and resource orchestration (Kohtamäki et al., 2025; Verbeke et al., 2011).

In sum, servitization requires a radical change in the salesperson's knowledge, logic and behaviour (Tuli et al., 2007). Communication capacity, made more difficult by the intangible nature of services, must be strengthened (Favoretto et al., 2022; Koponen et al., 2019; Zhang & Banerji, 2017). Aids come from the adoption of models to quantify the costs and benefits of the service along the life cycle (Raja et al., 2020; Rapaccini, 2015), as well as by adopting the concept of customer lifetime value (Eggert et al., 2020; Ulaga & Eggert, 2006). Being capable to use configurators, simulators, and demonstrators is crucial as well (Fischer et al., 2023; Kohtamäki et al., 2025; Ritter & Pedersen, 2020). As a result, the skillset of salespeople in servitizing firms must encompass not only technical and commercial expertise, but also abilities for value-driven communication, cultural sensitivity, leadership, and relationship management (Favoretto et al., 2022; Kowalkowski et al., 2025; Sklyar et al., 2025). Consequently, the gaps can be considerable, typically addressed through intense training programs (Kindström et al., 2015). From a theoretical standpoint, these transformations map into the micro level of Kozlowski & Klein's framework (Kozlowski & Klein, 2000).

2.5. Synthesis of extant literature and paper positioning

To clarify the conceptual positioning of the present study, Table 1 summarises prior research on the transformation of sales functions in the move to advanced services, structuring the relevant findings according to the dimensions of the multilevel framework. For each contribution, the same table provides its conceptual foundation, the methodological approach and the key findings. The table concludes by outlining the positioning of this study in relation to the existing literature.

Table 1 reflects the infancy and heterogeneity of the knowledge base on selling and sales management in servitization. A substantial proportion of extant studies are reviews or conceptual papers (e.g., Favoretto et al., 2022; Kowalkowski et al., 2017; Witell et al., 2020; Zhang & Banerji, 2017), complemented by a few empirical works (e.g., Rapaccini et al., 2023; Storbacka et al., 2013). While these strands have advanced understanding of structural redesign as well as of salesforce training needs, the findings have limits in connecting these different layers (Kowalkowski et al., 2025). Furthermore, the current landscape underscores the need for additional empirical research. This study addresses these lacks by offering a comparative, multilevel analysis that explicates how organisational structures (macro), processes and tools (*meso*), and knowledge and skills (micro) are reconfigured in tandem when base and advanced service logics coexist, thereby clarifying where this work extends the existing literature as mapped in Table 1.

3. Methodology

3.1. Research strategy and case selection

This study, conducted between January 2024 and March 2025, adopts a multi-case research strategy (Eisenhardt, 1989; Robson, 2002). This approach is particularly suitable for exploratory and explanatory research (Eisenhardt & Graebner, 2007), to answer questions such as *why*, *what,* and *how* (Yin, 2009), and to allow a rich and holistic understanding of the dynamics at work when boundaries between the investigated phenomena and the context are blurred (Meredith, 1998). The unit of analysis of our cases is represented by the sales functions in manufacturing firms undergoing servitization. Each case study has at least two built-in units, that is, the function/teams selling base services (i.e., minimum level of servitization), and the function/teams selling advanced services. This articulation makes it possible to distinctly analyse the logics that regulate different levels of complexity. In terms of temporal design, the study adopts a cross-sectional approach (Bryman, 2006): data collection was conducted at a single point in time within each case study. This design aligns with a comparative research strategy, and it is suitable for examining organisational variation across firms positioned at different stages of servitization (Bryman, 2006). In particular, the study contrasts two polar cases (Eisenhardt & Graebner, 2007)—each representing a distinct point along the Baines staircase (Baines et al., 2024)—to identify macro, *meso*, and micro-level differences in how the sales function supports the transition from base to advanced services. Case studies were purposively selected among over 40 companies affiliated to an Inter-University Research Centre on Servitization. Drawing on well-established knowledge and connections developed in several years of collaborations, the authors contacted executives of firms meeting the preliminary screening criteria.

Based on their availability and alignment with the research objectives, 5 firms were selected. The criteria used included: a) different proportions from the sale of base and advanced services; b) different configurations of sales functions and sales teams; c) different extents of digitisation of sales practices. Purposive sampling ensured variation in contextual conditions, consistent with the logic of theoretical sampling in qualitative research (Eisenhardt & Graebner, 2007). The 5 selected companies are multinational firms with global presence, operating in distinct industrial sectors including energy, electrical and power management equipment, industrial automation, packaging machines for

Table 1
Summary of prior research on the transformation of sales functions in servitizing firms, and positioning of this study.

Level	Reference	Conceptual foundation	Methodology	Key Findings
Macro	Storbacka et al. (2013)	Solution selling requires interdependent changes in customer orientation, organisational network, and operational flexibility.	Conceptual framework with qualitative fieldwork	Demonstrates that the required changes in organisational networkedness and operational adaptiveness for solution business models, particularly within sales functions, differ significantly between different business logics, impacting their organisation, roles, and strategic approach.
	Kindström et al. (2015)	Service infusion demands a shift in managerial control and sales organisation design to support value co-creation.	Qualitative case studies (5 firms)	Shows that service infusion entails increasing salesperson proactivity, adjusting managerial control mechanisms, and enhancing cross-functional interfaces to support customer-centric value creation.
	Kowalkowski et al. (2017)	Servitization as a shift in organisational identity and governance requiring realignment between product- and service-oriented logics.	Literature review	Highlights how misalignment between product- and service-oriented logics can create organisational tensions, and stresses the importance of leadership and governance alignment to support servitization trajectories.
	Zhang and Banerji (2017)	Organisational-level challenges as structural barriers to servitization outcomes.	Systematic review (48 articles), citation analysis	Proposes five organisational-level challenges—structure, business model, development processes, customer management, and risk management—and formulates hypotheses on how they hinder benefit realisation, calling for structural adaptations to support servitization.
	Favoretto et al. (2022)	Digital servitization as a multidimensional transformation, involving structural redesign and cross-functional coordination.	Systematic literature review (180 articles), NVivo-based content analysis	Identifies nine interconnected dimensions of digital servitization, including organisational structure, role redefinition, and internal coordination mechanisms, highlighting their interplay in enabling effective transformation.
	Rabetino et al. (2018)	Operationalisation of servitization through five enablers: digitalization, integration, capabilities, processes, and alignment.	Bibliometric and qualitative review (>1000 articles)	Identifies five foundational enablers—digitalisation, customer integration, service delivery processes, dynamic capabilities, and organisational alignment—that facilitate the operationalisation and scaling of service offerings across technology- and process-driven environments.
	Kohtamäki et al. (2020)	Digitalization moderates the link between service implementation and sales performance.	Quantitative analysis (Stata, ROA growth)	Finds that the interplay between digitalisation and servitisation positively moderates the relationship between service business implementation and sales performance, highlighting the importance of process design alignment.
Meso	Witell et al. (2020)	Sales function evolve in 4 distinct phases to sell advanced services	Conceptual and exploratory paper based on a theoretical synthesis of existing literature	Reveals that advanced service sales require alignment of value co-creation activities across multiple phases, and identifies distinct roles and resources needed in each stage.
	Rapaccini et al. (2023)	Modularity and digital platforms enable scalable yet customizable service solutions.	Longitudinal case study (2018–2021)	Reveals how modular digital platforms and external consulting practices co-evolve with client processes, enabling the orchestration of repeatable yet customised service configurations through adaptive role enactment.
	Sklyar et al. (2025)	Routinised digital practices as configurations enabling adaptive service delivery.	Fuzzy-set Qualitative Comparative Analysis on survey data from 112 manufacturing firms.	Identifies multiple configurations of routinised behaviours (assertive, agile, and reactive) within sales and service units, showing how these enable adaptive, process-aligned service delivery across digitally enabled ecosystems.
	Tuli et al. (2007)	Salespeople as co-creators of customer solutions, engaged throughout the life cycle.	Discovery-oriented interviews	Demonstrates that salespeople are expected to contribute beyond transactional exchanges by supporting the implementation, integration, and ongoing use of solutions, thereby requiring new relational skills and active engagement in customer post-sale processes.
	Ulaga and Reinartz (2011)	Hybrid offerings require new mental models and customer-centred value framing.	Case-based and conceptual framework	Highlights that transitioning to hybrid offerings requires salespeople to adopt a value-in-use logic, which necessitates new consultative capabilities, a shift in cognitive frames, and the development of customer-centred problem-solving skills to effectively communicate complex value propositions.
Micro	Terho et al. (2012)	Value-Based Selling (VBS) as a strategic approach that reshapes salesperson cognition and interaction logics.	Case studies (two B2B firms), qualitative data	VBS requires salespeople to develop a deep understanding of customer operations, quantify and articulate value in monetary terms, and engage in collaborative dialogue to co-create solutions that align with the customer's strategic objectives.
	Ulaga and Loveland (2014)	Hybrid selling requires a shift from transactional to consultative and relational roles.	Qualitative synthesis (interviews, focus groups)	Finds that service-oriented selling requires the development of enduring relational and problem-solving skills, as well as a redefinition of the salesperson's role to act as a strategic advisor and trusted partner in co-creating value within complex service interactions.

(*continued on next page*)

Table 1 (*continued*)

Level	Reference	Conceptual foundation	Methodology	Key Findings
	Salonen et al. (2021)	Adoption of VBS depends on salesperson-level traits and digital readiness.	Fuzzy-set QCA, survey data	Shows that the successful adoption of value-based selling requires a specific configuration of individual-level attributes, particularly adaptability, technology foresight, and customer orientation, which enable salespeople to align with the demands of digitally enabled service business models.
Multilevel	This study	Sales transformation as a multilevel and configurational process is shaped by the interplay between roles and governance ambidexterity, modularity, and hybrid digital enablement.	Comparative case study with polar approach of 2 global manufacturing firms at different stages of servitization	Develops a theoretical framework of ambidextrous sales configurations, highlighting how organisational, processual, and individual-level elements align differently in base vs. advanced service sales. Shows how firms orchestrate internal autonomy, organisational modularity, and digital tools to manage different sales logics under one roof.

food production, and professional laundry equipment. From these five, we retained two polar cases to maximise analytic contrast on our focal constructs while preserving contextual comparability and data depth for a rigorous multilevel analysis. In line with the Baines et al. (2024) staircase, the retained cases anchor opposite ends: one advanced-services-oriented and one base/intermediate-oriented, with both firms supplying complex, business-critical industrial equipment and granting information-rich access (multiple informants, internal documentation, archival sources). Manufacturers not retained lacked, in combination, sufficient polar distance on the Baines staircase, the multilevel access required for robust triangulation, and like-for-like industry comparability.

3.2. Data collection

Data collection was conducted within European subsidiaries of the five selected multinational firms and started with 12 semi-structured interviews with company managers. The interview protocol, used to screen and characterise candidate firms, covered four domains: (i) the service portfolio (base, intermediate and advanced), revenue mix and exposure to outcome- and performance-based agreements; (ii) the configuration and governance of the sales function (structures and roles, responsibilities and incentives); (iii) sales processes, tools and the degree of digitisation (e.g., CRM or other service-management platforms integrated with remote monitoring and analytics); and (iv) access conditions for triangulation (availability of multiple informants, internal documents and archival data). Illustrative prompts included: “Describe the composition of your service portfolio and the prevalence of outcome- or performance-based contracts”; “Outline structures and roles within the sales function and recent changes due to servitization”; “Which digital tools support pre-bid, negotiation, implementation, and delivery?”; “What documentation and data can be shared for triangulation?”

Afterwards, to strengthen the external validity of the results and verify their practical relevance, an open workshop was organised with over 25 managers from the participating companies. This event included an initial overview of the main trends in servitization by a noteworthy expert on the topic of service sales, followed by the authors' presentation of the preliminary results of this study. Participants included senior- and mid-level managers from departments such as service engineering, marketing, digital transformation, business development, and customer success. Participants were invited to provide critical observations and reflections, thus contributing to the collective validation of the research findings. Although not formally included in the final dataset, the insights from this exploratory phase informed the refinement of the research design, case selection, and development of the interview guide for the main case studies.

Based on the results of this exploratory phase, the most contrasting cases were used to further the study. This aligns with the polar-type approach proposed by Eisenhardt and Graebner (2007). In practice, this approach involves the purposive selection of cases at the extremes of a relevant variable (i.e., the extent of servitization) to facilitate the observation of contrasting patterns and to develop clear and robust theoretical propositions. It is also said that polar cases are ‘particularly suitable for illuminating and extending relationships and logic among constructs’ (Eisenhardt & Graebner, 2007, p.27), thus revealing unexpected dynamics. The selected polar cases share characteristics that make them particularly suitable for comparison: both operate on a global scale and provide complex industrial equipment that is critical for customers' business (Rapaccini & Visintin, 2015). As shown by Table 2, these firms were selected because they occupy different positions on the servitization staircase (Baines et al., 2024).

Case A mainly offers base services. Recently, thanks to a new organisational unit dedicated to servitization and digitisation, it has developed a centre for the remote control of connected plants, to develop the sale of intermediate and advanced services, such as remote monitoring, diagnostics, help desk and troubleshooting. The company has structured a portfolio of maintenance packages to become a strategic partner to their customers. However, the transformation is in its early stages. On the contrary, Case B offers all types of services and has developed servitization for a couple of decades. In this firm, base services still represent an important part of service turnover, but profits essentially originate from long-term full-risk service agreements, through which the company guarantees the availability of its machines. We conducted 10 semi-structured interviews, involving interviewees with heterogeneous roles. The interviews, lasting between 62 and 113 min (see Table 3), were conducted in person or through secure digital platforms, recorded and transcribed in full. The structure of the interview guide was informed by the multilevel theoretical framework (Kozlowski & Klein, 2000) introduced earlier—namely (a) organisation,

Table 2
Description of polar cases.

Case	Description and size	Servitization stage
A	A European multinational company active in the production of industrial equipment for house furniture fabrication. It operates on a global scale, with facilities in Europe, North America and Asia. The company has a global turnover of 1bn euros and 4 k employees.	The base services business has been consolidated, the development of intermediate services has been initiated, and some technological enablers have already been developed. The long-term goal is to develop advanced services, but presently, the services business accounts for less than 25% of total turnover and mainly comprises base services.
B	A division of a global industrial company, based in Europe, producing technologies for oil & gas, energy, and utility industries. It provides advanced solutions for remote control and diagnostics to a large fleet of connected machines. Its global turnover is around 2.3 bn Euro with 5.5 k employees	The base, intermediate, and advanced services businesses are consolidated. The sales of services generate over 50% of total revenues, and advanced services generate the largest shares of profit.

Table 3
Interviewed managers of polar cases.

Case	Role	ID	Interview length [min]
A	Head of Digital Transformation and Business Remodelling	Int1	113
	Global after-sales (parts & services) manager and business development	Int2	76
	Senior Service Director	Int3	85
	Digital Projects Manager	Int4	62
	Global Marketing Manager	Int5	67
	Digital Product Management Executive	Int6	103
B	Global Service Sales Director	Int7	72
	IT Director - Global Service Repairs	Int8	63
	Commercial Operations Manager	Int9	101
	Services Growth and Strategy Director	Int10	112

roles and responsibilities (macro level), (b) processes, tools and technologies (*meso* level), and (c) knowledge and skills (micro level)—which provided a preliminary deductive lens for exploring the evolution of the sales function under servitization. However, the research followed a flexible and iterative approach, allowing for inductive refinement as data collection progressed. As interviews unfolded, this openness enabled the emergence of cross-level dynamics and alignment issues across macro, meso, and micro domains, which became central to the theoretical framework developed.

3.3. Data analysis and interpretation

Data analysis was explicitly undertaken in line with the multilevel theoretical framework of Kozlowski and Klein (2000). The transcripts were analysed with a mixed coding procedure, combining deductive (based on theoretical concepts) and inductive (data-emergent) approaches. The analytical dimensions identified in the literature review—that is, macro (organisational structures, roles and responsibilities), meso (processes and tools), and micro (knowledge and skills) levels—initially informed the construction of the interview guide and deductive codes. However, the multilevel misalignments and dynamic interactions across levels emerged inductively during analysis and led to the development of the final theoretical framework. The empirical corpus was completed by direct observation activities at the commercial offices, combined with the analysis of secondary data sources such as organisational charts, training materials, and sales presentations. The researchers analysed the data collected in two moments: an intra-case analysis aimed at building detailed narratives of the observed transformations, and a cross-case analysis aimed at identifying recurring patterns and structural differences in the sales function, moving from lower to higher levels of servitization. In reporting the results, Section 4 is organised thematically (macro, meso, micro), consistent with the adopted theoretical lens (Kozlowski & Klein, 2000), to contrast the two polar cases to foreground key differences and avoid duplication that would arise from separate within-case narratives. Section 5 then offers the integrative cross-case synthesis and the emergent framework.

Finally, the insights were discussed in a workshop attended by representatives of the companies involved in the case studies, as well as by other managers of manufacturing firms undergoing servitization. The workshop was organised to validate the paper's results and enhance the transfer of knowledge generated by this study. Table 4 summarises the research stages of this study.

4. Findings

This section presents the empirical findings emerging from the polar cases. The structure mirrors the multilevel framework outlined in Section 2, with results organised around the corresponding levels of change, respectively a) organisational structures (macro), b) processes and tools (meso), and c) knowledge and skills of salespeople (micro).

Table 4
Research stages.

Stage	Activities	Outputs
Preliminary exploration	Purposive selection of 5 manufacturing firms. 12 semi-structured interviews with sales and service managers.	Mapping of organisational configurations of services sales functions. Identification of initial patterns related to servitization levels.
External validation and case selection	Open workshop with senior managers from 5 selected firms. Presentation of preliminary findings and collection of feedback.	Validation of early evidence. Selection of two polar-type cases (higher vs lower servitization).
In-depth case study analysis	10 semi-structured interviews. Document analysis, observation, and secondary data review. Mixed coding (deductive + inductive).	Within-case narratives of organisational change. Cross-case comparison of sales functions at different servitization stages.
Final validation	Concluding workshop, also with managers from other companies. Discussion of key propositions and insights.	Validation of theoretical contributions. Knowledge transfer. Shaping of managerial implications.

Within each subsection, the findings are presented by describing the two polar cases in relation to each level of analysis, enabling contrasts and similarities to emerge. Then, a cross-case comparison is developed in Section 5.

4.1. Organisational structure

In both the polar cases, the service sales teams operate within the company's service business units and are therefore separate from the product sales teams. Beyond this similarity, the cases differ in their organisational structures. These differences reflect their respective stages of servitization and lead to different sales governance mechanisms. The service sales organisation of case A is structured around two main teams, respectively, the Service Key Account Managers (SKAM) and the Service Back Office Team (SBOT). SKAM oversees managing strategic customers, that is, those who show high purchase potential. The goal is to develop a strong partnership with them. As previously noted, this company has recently developed its offer of maintenance packages, which include training for equipment operators. The goal is to ensure that the potential of the supplied equipment is fully exploited and that customer value is maximised. The primary responsibility of SKAM is to help important customers in identifying the maintenance package that better fits their needs and then to convince them to buy it. Selling a multi-year maintenance contract is considered a prerequisite to the development of advanced services. However, customers must pick their package from a rather standardised and limited offer, and sellers have no autonomy for either customising the terms and conditions of contracts or negotiating their prices. In this situation, SBOT oversee the selling of base services to non-strategic customers. Alongside these two main figures, the sales function leverages a network of internal technical enablers—referred to internally as *influencers*—who maintain frequent customer touchpoints across the service journey (Witell et al., 2020). These professionals are not directly involved in selling but can facilitate the work of both SKAM and SBOT. Their contributions are twofold: they reveal opportunities for service sales and provide technical knowledge to the sales team. Influencers include senior technicians of installation and help desk departments, area managers, and service managers. The approach to service sales in case A is typically proactive. As pointed out by one manager interviewed:

"*We never wait for the customer to call us. We are always the ones who take the initiative, proposing solutions that can anticipate their needs, whether it is spare parts to be kept in stock at the customer premises, or a maintenance contract that reduces the risk of shutdowns*." (Int2).

Although SKAMs are expected to play a strategic role in servitization, their scope of action is constrained by strict standardisation logics and the limited autonomy granted by top management. This reveals a structural tension between the formal strategic ambitions of the firm and the actual organisational enablers allocated to the sales team (Kindström & Kowalkowski, 2014; Tuli et al., 2007). As one manager observed:

"We work with standard modules, but we have to know how to put them together like a puzzle. We can't change the pieces, but we can choose the order, the time and how to present them." (Int5).

Standardisation and modularity were intended to simplify sales decisions, mitigating the complexity and fostering the *go-to-market* of the service business worldwide. In line with the literature on service modularity (Brax et al., 2017), this approach enables replication and scalability—but may also inhibit contextual adaptation. Above all, this modularity has been designed to drive the evolution of the sales function gradually: salespeople were expected to learn over time from this predefined framework, before being given the responsibility of handling more complex businesses. As underlined by a senior manager:

"Business remodelling and strategic evolution go faster than people can adapt. [...] We could potentially offer more than the three current service packages, but we must acknowledge and navigate the human realities that characterise our organisational context." (Int1).

This modularity ends up constraining the creativity of the sales team, generating a discontinuity between theory and practice. On the one hand, the strategy recognises that vendors play a leading role in the transition to advanced services. On the other hand, the other company's directions treat them as mere executors of predefined offers. This creates a misalignment between formal structures and the skills and behavioural requirements for consultative selling (Ulaga & Loveland, 2014; Raja et al., 2020). The leadership believes that it is necessary to contain their freedom, which could generate numerous inconsistencies and inefficiencies. Modularity in case A is therefore used as a control mechanism, as well illustrated by the Service Director:

"The modularity of services arises from the need to harness the extreme creativity of salespeople [...]. They only have to say A, B, C."(Int3).

While modularity is conceptually linked to process optimisation and coherence (Cenamor et al., 2017; Sjödin et al., 2019), its managerial deployment here appears to conflict with the need for contextual sensitivity and salesperson empowerment—both of which are central to the effective sale of advanced services. The sales teams don't seem ready to grasp business opportunities and transform them into actionable propositions. The Head of Business Development sums up this frustration as follows:

"What we expect from our salespeople today is that they can raise customer awareness of advanced services [...]. But to do that, they also need to have the space to move around the company." (Int2).

In the absence of this legitimacy, the sales organisation remains disconnected from the core processes of the firm and cannot influence its service innovation. From a theoretical standpoint, these tensions illustrate a macro-level misalignment between formal roles, governance mechanisms, and expected strategic skills (Kozlowski & Klein, 2000).

In summary, the sales function of case A appears pressured by unresolved tensions between autonomy and delegation, and between replicability and adaptability. The modularity of the offer, while representing a necessary condition to enable servitization, seems not sufficient to culturally and strategically transform the sales function and the role that it exposes to other departments. This finding resonates with prior research indicating that overly rigid modular configurations, although beneficial for internal efficiency, may hinder the relational and consultative behaviours necessary in advanced service sales (Brax et al., 2017; Sklyar et al., 2025). Such configurations risk reinforcing product-centric routines rather than enabling the adaptive capabilities required for servitization (Kohtamäki et al., 2025; Tuli et al., 2007).

In case B, the sales function is organised into specialised teams. Each team is solely responsible for selling a specific type of service. The main distinction is made between base and advanced services. The sales function of base services is structured into three operating units: the Service Back Office Team (SBOT), the Service Operations Excellence Team (SOET), and the Professional Services Sales Team (PSST). As in case A, the SBOT deals with the sale of parts and repairs, that is, it tries to provide a fast reply to the numerous requests for quotation that come from customers. The SOET give support, coordination and analysis. They provide technical support and give information about inventories and delivery times of on-hand orders, estimate shipment cost and time for parts not in stock. Doing this, they help in responding to RfQ with technical expertise. Finally, the PSST is dedicated to managing non-routine requests, such as requests for spare parts and repairs of obsolete components and more complex situations that require a work-around. In this, they often collaborate with other business functions. This team has greater autonomy to customise the offer in terms of prices and delivery options. In contrast, in advanced services, the sales function adopts a diametrically opposite logic. The sale of advanced services is managed by a selected group of experts who operate with a consultative approach. As such, the team is highly skilled and responsible for pursuing global sales. The professionals in the team participate in trade fairs, organise customer events to reinforce the relationships with them. This is well represented by what was told by a manager:

"*Contractual service agreements are our flagship, an atelier run by a few experts operating on a global scale. [...] By working proactively, they seek to create timely relationships with customers to sell them customised solutions that meet every need*." (Int10).

This sales team has three main responsibilities: a) active promotion of the company's reputation and capabilities to deliver customer outcomes, b) creation of touchpoints for structured discussions with the decision-making units of the customer organisations, and c) the possibility of personalising long-term contractual relationships. These responsibilities are in the hands of an agile team, which includes senior project managers who master the technical knowledge around industrial equipment and production processes. In addition, the firm has developed over time advanced analytics tools and predictive capabilities, built on IoT data from connected machines. This sales team has just one business logic: being the partner that guarantees the highest availability and productivity of the supplied assets, under agreed contractual terms. In case B, this logic is supported by a Steering Committee, an inter-functional unit which is responsible for solving ambiguities and tensions between independent sales teams. The Steering Committee determines whether a specific customer should receive an offer of base services or whether it is strategic to propose advanced services and long-term agreements. Doing this, the Steering Committee mitigates conflicts from uncoordinated or inconsistent commercial actions. As stated by one manager:

"*The Steering Committee was created to overcome internal competition between the various sales units. Now we guarantee a consistent and unambiguous answer to the customer*." (Int7).

For the same reasons, the company's sales leadership is unified to coordinate the different product and service sales units. This leadership has profoundly revised the incentive mechanisms for service sales, introducing criteria that reward collaboration rather than one-shot deals. It follows that the sales governance of case B aims at mitigating conflicts from fragmented responsibilities and misalignment, thus enhancing consistency and enabling faster and more effective decision-

making about delivering customer proposals. This organisational architecture reflects a more advanced configuration of macro-level change, where structural alignment, authority distribution, and inter-functional coordination mechanisms are designed to support a service-dominant logic (Gebauer et al., 2017; Kohtamäki et al., 2020). Unlike in case A, modularity in B is not used to simplify the offer or to limit the discretion of sellers, but as a tool to manage the complexity generated by this high customisation. The modules, which in A are used to contain and standardise, in B constitute an architecture that can be flexibly and quickly adapted to build tailor-made solutions. This approach requires salespeople to be capable of independently assessing the technical and economic feasibility of each customisation, acting authoritatively toward both the customer and internal functions. The level of autonomy and cross-functional legitimacy these vendors enjoy is supported by digital tools. Access to data generated by the fleet of connected machines, insights into product and service performance, as well as the application of simulation models, all of this enables the development of credible value propositions that are aligned with the company's operational capabilities and strategic targets. In this sense, the PSST operates in the form of solution architects, which are typical of the digital industry. These are capable of interacting with heterogeneous interfaces internally and toward the customer's organisation, along the relationship life cycle. This more dynamic interpretation of modularity aligns with recent literature that positions modular architectures as enablers of cross-functional integration and strategic flexibility in service-oriented firms (Favoretto et al., 2022; Rapaccini et al., 2023). The presence of governance mechanisms such as the Steering Committee echoes calls for institutional arrangements that mitigate coordination failures across differentiated service logics (Kowalkowski et al., 2017; Rabetino et al., 2018).

In sum, the polar cases exemplify contrasting macro-level configurations: while Case A constrains sales autonomy through rigid standardisation and weak cross-functional integration, Case B leverages modularity, governance, and coordination mechanisms to empower sales teams as strategic actors of the servitization journey. This divergence highlights how structural enablers can either inhibit or foster the capabilities required for advanced service selling, anticipating tensions further examined at the *meso* and micro levels.

4.2. Processes and tools

The two cases also show different approaches in structuring the sales process. In case A, the sale is configured as a rigidly codified process. The activities are organised according to a **blueprint** that specifies the actions to carry out, distinguishing between primary and secondary actors. This also reflects the little autonomy **of the service sales function**. This reflects a logic of standardisation and control aimed at ensuring efficiency and replicability, typical of early-stage servitization (Kindström & Kowalkowski, 2014; Kohtamäki et al., 2020). The sale process develops through a 3-phase customer journey: a) *Offering*, in which salespersons present the maintenance package suggested for each customer segments; b) *Startup*: in which the salesperson supervises the start-up of service delivery, ensuring alignment with established standards; c) *Operations*, in which the seller oversees the relationship with the customer and supports continuous improvements of the service delivery process. However, this phased logic tends to remain transactional rather than relational, limiting the opportunities for customer co-creation. While aligned with traditional product sales approaches, this sequential and blueprint-driven process does not fully capture the iterative and exploratory nature of advanced service selling (Reinartz & Ulaga, 2008; Witell et al., 2020). Our analysis also highlights some issues that prevent the development of advanced services. Some tools designed for product businesses do not effectively support the sale of advanced services. As highlighted by the servitization manager of the company:

> "*Our service model is changing faster than our tools. If we want to sell outcomes and not just products, we must put sellers in a position to act as value orchestrators, not as simple executors of a policy.*" (Int3).

In this context, the firm's technologies should also evolve to allow the sales function to play a key role in value co-creation.

In case B, the sales process of base and advanced services largely differs. In base services, the process is triggered by Request for Quotations, which are received from customers. This process is carried out massively. The sales team must respond very promptly, as productivity and response times are key drivers of competition. Despite the apparent simplicity of this model, its success depends largely on the adoption of digital tools for workflow management. The more these tasks are standardised, the more they can be codified and automated, the lower the response time. Our research shows a discontinuity between the available technologies and their use to support services sales. While advanced services benefit from breakthroughs such as IoT, machine learning, predictive analytics and simulators, base services are still managed through old-style tools such as spreadsheets and stand-alone databases that hinder the traceability of the progress of sales processes. This seems to limit the proactivity and proficiency of selling base services.

As one manager pointed out:

> *"We have monitoring systems for advanced services, but in base services, we still use Excel. We don't have a system that tells us what we sell to each customer or helps us predict when they will need a part."* (Int7).

The gap between the tools used for selling advanced services and those of base services is due to various factors, such as the organisational inertia and strong product-dominated culture of the base services consolidated business, the reluctance of customers to share sensitive data, and the difficulty of aggregating the numerous information sources. As another respondent noted:

> *"Our main problem is the lack of advanced tools that proactively monitor and analyse customer consumption and purchases of spares. If the salesperson doesn't notice a need or the customer doesn't call us, we remain in the dark."* (Int8).

From a process perspective, this reactive model illustrates the persistent tension between product-legacy practices and service-driven transformation—a theme largely overlooked in the servitization literature (Kohtamäki et al., 2020). Overall, case B highlights how the sales function in base services is still anchored to old practices, which limit its manoeuvrability in increasingly competitive markets. Conversely, the sale of advanced services is structured into three phases, respectively *offering, implementation* and *delivery,* which seems close to what was observed in case A. However, the logic is profoundly different, as the sales team of case B can largely tailor their proposals, and they are also involved in customer needs exploration (*demand to inquiry*) independently from the offering-to-delivery process. Once the customer requires a quotation for advanced services (*inquire to order),* the team collects the requirements, develops the proposal and supports the customer in issuing the purchasing order. Subsequently, in the *implementation phase*, the salesperson participates in solution configuration and implementation, coordinating the internal resources (as a project manager) and connecting the firm's technical teams to the customer organisation. As mentioned, at this stage, there is still room for customisation. During the *delivery phase*, the sales team continues to oversee the relationship with the customer, monitors the performance, collects feedback and identifies opportunities for selling additional services. This process is supported by dedicated tools. In the *demand to inquiry* phase, the sales team uses a specific Customer Relationship Management (CRM) system that suggests customer profiles and helps to identify business opportunities. The most important tool is, however, a simulator/configurator tool, which is used to assess the feasibility, delivery times and costs of a given solution. This largely reduces response times for elaborating the proposals and

enhances the perceived quality of the customer interactions. In the implementation phase, the team is supported by an integrated document sharing and project management system that facilitates cross-functional collaboration and ensures transparency and consistency of information flows. Finally, in the delivery phase, the sales team have some data analytics and specific dashboards that are built on the data collected in real time from the connected machines. They analyse the performance of the equipment and can therefore identify up-selling opportunities and suggest the need to revisit the contractual conditions. This comprehensive technology-enabled process resonates with recent conceptualisations of advanced service sales as a continuous, iterative, and multistakeholder-driven business (Kowalkowski et al., 2025; Witell et al., 2020). However, our findings suggest that such configurations remain rare and difficult to implement, given legacy systems and institutional resistance—thereby contributing new empirical insights to a literature that has often assumed process transformation as unproblematic.

4.3. Knowledge and skills of salespeople

Both cases A and B confirm that salespeople are subject to training and reskilling because of servitization. However, how such training is designed and implemented can significantly vary. In case A, training focuses on reinforcing standardised routines and service protocols related to maintenance packages. Salespeople are expected to acquire deep knowledge of predefined offerings and follow structured scripts during interactions. The goal is to transfer in-depth knowledge to adhere effectively to pre-established selling approaches. As highlighted by a function manager:

> "*Everyone involved has responsibility and knowledge for carrying out each specific activity [...] precisely with the expected productivity.*" (Int5).

This model presumes that performance stems from procedural compliance rather than commercial initiative. As another informant notes:

> "*The salesperson visits the plant, chats with the maintenance guy, stops by the purchasing office... It's the classic rep behaviour: you visit, you build relationships, maybe go to dinner… then you send the quote list to [the headquarters of the Company. But in the end, even though we stay close to the customer and proactively maintain contact, we mostly end up offering predefined packages — the room to propose tailored solutions is limited, so all this relationship-building sometimes feels constrained, almost wasted.*" (Int2).

Salespeople in case A thus act as executors of predefined business proposals, exhibiting a transactional logic that limits customer engagement. This echoes findings by Ulaga and Loveland (2014), who argue that traditional sales teams often lack the motivation and skills to shift from reactive to proactive engagement. Moreover, this approach contrasts with the value co-creation ethos required for advanced services, as it discourages anticipatory sensing and relational exploration. This low level of proactive involvement also reflects the organisational barriers to solution selling identified by Panagopoulos et al. (2017), who highlight how rigid sales routines and insufficient integration into solution development processes hinder salesperson effectiveness.

In contrast, case B fosters a more strategic and developmental logic. It proposes a radically different training concept, which is based on the development of cross-disciplinary skills, such as in consulting firms. Vendors of advanced services are selected on the basis of their motivation and experience in product engineering and project management. These are subsequently empowered by specific training programs, which develop both analytical capabilities (e.g., financial justification, risk assessment, technical feasibility) and soft skills (e.g., empathy, communication, relationship building) to cope with the complexity induced by servitization. Sales teams are also proficient in using digital tools such as configurators and simulators, using which they assess the feasibility and viability of innovative solutions. A manager of the company summarises these benefits as follows:

> "*The world of advanced services is our goose that lays golden eggs. We know how to do this well. We have built on it over time. We know our equipment in every detail, the performance, the vulnerabilities. This allows us to offer more than just maintenance. We know how to guarantee continuity, performance, and productivity. We don't just sell services, we sell peace of mind.*" (Int10).

This statement highlights how much the expertise of sales teams enables trust between the counterparts. Salespersons are thus empowered with essential skills to play a vital role in connecting the provider to its customers, and this is considered a strategic investment in the move to advanced services. This is consistent with the literature (e.g., Blocker et al., 2012; Terho et al., 2012), which highlights the need for solution-oriented skills in servitized contexts. Moreover, sellers are empowered to manage more complex stakeholder interactions within customer organisations—including IT, operations, and finance—requiring cultural fluency and negotiation acumen (Kowalkowski et al., 2025; Rapaccini et al., 2023).

In sum, the 2 cases show diverging interpretations of what constitutes a 'skill' in service sales. Case A prioritises procedural consistency and sales control, whereas case B cultivates the cognitive, relational, and digital competencies that support adaptive solution selling. From a theoretical standpoint, these findings align with the micro-level changes described in our conceptual framework (Kozlowski & Klein, 2000) and confirm prior research showing that only a fraction of legacy salespeople successfully transition to advanced service roles (Kowalkowski & Ulaga, 2017; Ulaga & Loveland, 2014).

5. Cross-case comparison

This section develops a cross-case comparison between the two polar cases. Building on the empirical findings presented in Section 4, the analysis identifies similarities, differences, and underlying mechanisms across cases in relation to organisational structure, processes and tools, and knowledge and skills.

5.1. Organisational structures

The research confirms that the strategic evolution of the sales function reflects the extent of servitization (Kindström & Kowalkowski, 2014). In particular, the teams selling base services have organisation and approaches that notably differ from those selling advanced services. The configuration of responsibilities, the level of autonomy for proposal customisation, and the intensity of cross-functional coordination are all impacted by the extent of servitization. A product sales culture still dominates case A, although they are steadily moving into a more service-oriented business. The service sales organisation has been adjusted with the introduction of specific professionals, and this can be viewed as a first attempt at this evolution. However, the organisational structure remains rigid, and the service offer is built on pre-configured packages that sales teams cannot personalise. Sellers, in addition, have little autonomy in setting up price negotiations. Despite this, the team is asked to play a strategic role in servitization, as they are expected to identify and engage prospects to promote the company's advanced service offerings. This confirms the firm's intentions of gradually reconfiguring its approach to sales.

Case B, on the other hand, is more experienced in the journey of servitization. In this case, the organisations of sales teams for base and advanced services are largely consolidated and different. In base services, the organisation maintains an articulated structure that operates massively and reactively, to reply to the large number of RfQ. The responsibility for selling advanced services is instead in the hands of a smaller team of global experts, whose mandate is the creation of

strategic partnerships with selected customers. This dichotomy is supported by multi-level governance, which includes a steering committee and a unified commercial leadership, in charge of solving conflicts and promoting coherence of the sales strategy. Three considerations emerge from this research that can be used to summarise the differences in the organisation of sales functions of base and advanced services:

1. Control and standardisation vs. autonomy and customisation: in case A the offer of base and intermediate services is standardised, and salespersons have no freedom to modify the simple catalogue of maintenance packages; in case B a dual logic prevails, in which the sales organisation can significantly tailor the offer of advanced services, and even that of more complex base services.
2. Leadership and mechanisms for coordination and incentives: Case B exhibits a more sophisticated governance structure than Case A. In both cases, sales leadership is unified; however, in Case B it is supported by a Steering Committee to ensure greater coherence in selling strategies. In this latter case, the incentive system has been substantially revised to promote cross-functional collaboration and mitigate internal competition.
3. Role of the sales function: In Case A, service sales are regarded primarily as operational support for the product business. In contrast, in Case B, the advanced services sales team is recognised as a strategic driver of servitization, exerting a profound influence on the overall growth and success of the business.

This last point deserves further study. The SKAMs of case A are located at the interface between strategy and operations, but their mandate is largely constrained. Although they are responsible for developing relationships with strategic clients to develop the service business, in practice they are modulating executors. They can choose *when* and *how* to offer a maintenance package, but they can neither change the sales script nor customise the contents of the offered packages. All this originates from the compromise of enabling the transformation without losing control over productivity and compliance. This generates a misalignment, as sellers are invested with transformative expectations but have neither the autonomy nor the capabilities to fully realise them. Modularity, conceived as a servitization booster, also ends up acting as a tool for containing unregulated commercial initiatives. The lack of internal legitimacy – that is, the impossibility for the seller to activate transversal resources or influence cross-functional decision-making processes – further compromises the possibility for sales teams to act as co-creators of value. In contrast, the sales function for advanced services of case B is shaped as a highly professionalised and consultative unit, capable of finding innovative solutions to customer problems. Salespeople are hybrid figures, having a technical background which has been empowered with communication and people skills, organisational legitimacy (i.e. leadership), and business knowledge. This allows them to act as solution architects, who are free to engage resources and negotiate their value internally and externally. This does not appear to be the result of a precise top-down and early-stage organisational design, but derives from over time adaptation, due to a progressive reconfiguration of the service offer to capture and satisfy emergent needs. In this context, modularity is the base for personalisation, which is achieved with the use of sophisticated tools and well-structured cross-functional collaboration. In base services, however, the sales organisation of case B shows a persistent segmentation and remains anchored to traditional approaches. Sales teams thus continue to operate as agents prompted by customer requests. This represents a dissonancy of the overall maturity of the organisation, suggesting cultural rather than structural resistance: base services in highly servitized firms are perceived of secondary importance.

Based on this consideration, this study shows that it is necessary to pay particular attention to two critical issues when delimiting the strategic transformation of the sales function in servitization:

1. The divergence between the formal and substantial role of sales teams: in case A, the seller is formally at the centre of the strategic transformation but operates in a context that limits its autonomy and legitimacy. In case B, at least for advanced services, this coherence has been achieved through adaptation and experiential construction of this strategic role.
2. Modularity as a booster or constraint for developing servitization: In case A, modularity is a tool for productivity and compliance, while in case B, it provides the bricks to creatively build technical and commercial dialogues with both customers and internal departments.

5.2. Processes and tools

Sales teams of servitizing firms adopt practices and digital tools that differ between base and advanced services. In case A, sales processes are standardised to better control the productivity and efficacy of sales representatives. Tasks are extremely codified to ensure compliance with established policies and are supported by general-purpose tools. This constitutes a significant barrier to furthering the transformation. In case B, the sales approach of base services differs from that of advanced services. The former is typically reactive, being triggered by a request for proposal that must be prompted rapidly; otherwise, customers will likely buy parts and repairs from local competitors. Therefore, sales tasks of base services are continuously streamlined to increase their productivity. However, these sales are supported by general-purpose technologies, and this limits their traceability and improvement. The sales approach of advanced services is fundamentally different. The process embraces the entire customer life cycle and integrates practices that are typical of consulting firms, such as a clear focus on demand generation, followed by elaboration of personalised project-based offers. Vendors are involved in all these phases, up to implementation and delivery. Their job is facilitated by sophisticated tools and platforms for collaboration, customer relationship management, performance simulation, and technical and cost assessment. Advanced technologies for service innovation also enable sales scalability and data-driven decisions. The research sheds light on two critical issues to be considered in introducing processes and digital tools for selling advanced services.

1. Rigidity vs. adaptivity: case A embodies procedural views of the sales process, while case B – at least for advanced services – adopts more adaptive, flexible approaches that can more effectively capture contributions from outside and inside the organisation.
2. Digital tools as impediments or enablers: case A shows the limitations of general-purpose digital tools that hinder the evolution of servitization; case B shows that dedicated digital tools act as enablers for advanced services sales.

5.3. Knowledge and skills of salespeople

The research highlights how the level of servitization profoundly affects the knowledge and skills of salespeople required for the sales function. While in case A, training of sellers is functional to adherence to the standardised operating model, in case B, empowering the skillset of salespeople is key for furthering the advanced service business. It follows that the training programs of salesperson in base services can be delivered as compliance-oriented programs, with emphasis on the transfer of procedural knowledge and full adherence to codified business practices. Training is thus focused on the selling tasks, ensuring that each salesperson knows exactly *when* and *how* they are expected to execute their operations. Competence is not associated with the ability of grasping novel customer needs or designing tailor-made offers, but with mastering sales scripts. While effective in the initial stage of the servitization journey, this approach may constrain the move toward advanced services. In case B, on the contrary, training is intended to empower people and promote growth. Salespeople, especially in advanced services, are selected for their technical skills and

subsequently trained to complement these skills with managerial knowledge. This allows them to act with high autonomy and authority throughout the sales process, from demand generation to solution configuration and implementation. The training is highly personalised, oriented to develop these hybrid technical-commercial abilities. The use of advanced tools such as CRM, dashboards, configurators and simulators further enhances their abilities. In summary, the research highlighted three criteria for determining how the knowledge and skills of sales teams in servitizing companies should be empowered:

1. Compliance vs. empowerment: case A uses training approaches that ensures compliance and productivity of commercial tasks; case B instead empowers people to develop judgment, responsibility and business competence in salespeople.
2. Vertical vs. horizontal skillset: while case A shows a clear orientation on mastering procedural knowledge for sales execution, case B shows a focus on a more transversal skillset, which allows the evolution of salespersons toward consultative teams.
3. Static vs. adaptive training: case A has a rather static training model, which is easily replicable; case B adopts adaptive training programs, built around individual assessment and adapted according to the evolution of responsibilities.

Regardless of the training strategy, developing knowledge and skills of sales teams for servitization requires considerable efforts along the entire journey, as this change is progressive and must be persistently supported by the sales leadership and by adequate rewards.

6. A multilevel framework for changes to the sales function in servitizing firms

The cross-case analysis carried out in the previous section reveals that servitization requires a profound reconfiguration of the sales function, unfolding across multiple levels. This section builds on the cross-case comparison to develop an integrative framework that synthesises the empirical findings and explains how sales transformation unfolds during the transition from base to advanced services. In doing so, it provides a theoretical explanation of how alignment and misalignment across organisational levels shape both the evolution of the sales function and the tensions that arise when base and advanced services coexist. While prior studies have examined individual skillsets (e.g., Ulaga & Loveland, 2014), practices (Tuli et al., 2007), and organisational arrangements (Kohtamäki et al., 2020), this study proposes an integrated framework that conceptualises the transformation of the sales function in servitizing firms as a multilevel and dynamic process, mediated by alignment across macro-, *meso*-, and micro-levels of the business organisation (Kozlowski & Klein, 2000).

Building on the empirical evidence from the polar cases, the framework clarifies how the transition from base to advanced services reshapes the sales function across organisational structures (macro level), processes and tools (meso level), and knowledge and skills (micro level). In line with RQ(a), the analysis shows that change at any single level is insufficient to support the transition toward advanced services. Instead, effective sales transformation depends on the degree of alignment across these interdependent levels.

The findings further show that misalignments across levels give rise to persistent organisational tensions when base and advanced service offerings coexist within the same sales department. In line with RQ(b), these tensions emerge, for example, between control and autonomy, standardisation vs. customisation, and rigidity vs. flexibility, as documented in Sections 4 and 5. Such tensions are particularly evident in Case A, where strategic aspirations toward advanced services are not matched by corresponding changes in sales governance and practices, thereby constraining the evolution and effectiveness of the sales function. Conversely, Case B illustrates how higher levels of alignment across organisational levels mitigate these tensions and enable more responsive and relational forms of selling. In this case, integrative mechanisms—such as differentiated mandates, flexible approaches to modularity, dedicated digital tools, ad-hoc governance structures, and adaptive training—support the coordinated evolution of structures, processes, and skills required for advanced service selling.

Fig. 1 synthesises these insights by translating the empirical findings from Sections 4 and 5 into a multilevel framework. In particular, the elements reported under the heading 'Servitization extent' summarise key empirical characteristics that distinguish the polar cases—such as the degree of control vs. standardisation—which have been discussed in detail across the macro-, meso-, and micro-level analyses (Sections 4.1–4.3) and consolidated in the cross-case comparison (Sections 5.1–5.3).

The framework maps the three organisational levels, identifies key transformation domains at macro, meso and micro levels, and highlights the main tensions encountered along the path to alignment. Moreover, the framework differentiates between low and high alignment configurations, illustrating how multilevel coherence fosters the development of the advanced services business. This framework builds upon and extends the existing literature by conceptualising sales transformation as a recursive, multilevel alignment process—where shifts in structural arrangements (macro level), operational routines (meso level) and individual capabilities (micro level) co-evolve dynamically through continuous feedback and adaptation. Previous research has made important strides in delineating the requirements for successful service selling—highlighting the evolution of individual skills (Ulaga & Loveland, 2014), the extension of customer interaction processes (Tuli et al., 2007), and the role of organisational structures and digital tools (Kohtamäki et al., 2020; Sklyar et al., 2025). However, these dimensions have often been analysed in isolation or framed as static capabilities to be acquired. By contrast, our findings reveal that sales transformation in servitization is not a linear or modular progression, but rather a multilevel realignment process in which organisation, processes, and skills must co-evolve through iterative feedback and adaptation. This perspective deepens our understanding of the service transformation by framing it as a process of organisational learning, where the degree of alignment between different levels of the firm determines whether the transformation succeeds or stalls. The tensions observed between standardisation and personalisation of the service offering, or between control and autonomy of the sales teams, are not residual inefficiencies but rather structural tensions that reflect deeper misalignments between strategic intent and operational capacity. In doing so, the framework speaks directly to recent scholarly calls for more context-sensitive and temporally grounded explanations of how sales organisations navigate hybrid business models and transition across maturity stages (Kowalkowski et al., 2025; Raddats et al., 2019).

7. Conclusion

This study contributes to the literature on industrial sales and servitization by offering a theoretically grounded and empirically informed understanding of how the sales function transforms when base and advanced services coexist. Building on a multilevel framework, the research integrates macro, meso, and micro levels to explain the complexity of sales transformation in servitized manufacturing firms.

7.1. Theoretical implications

This study makes three theoretical contributions to the literature on servitization and sales transformation.

First, the study proposes a multilevel conceptualisation of sales transformation as a dynamic alignment process involving interdependent changes in organisational configuration, sales processes and tools, and the skills and knowledge of salespeople. While prior studies have explored these dimensions individually—focusing on structural separation (Kindström et al., 2015; Kohtamäki et al., 2020), extended sales

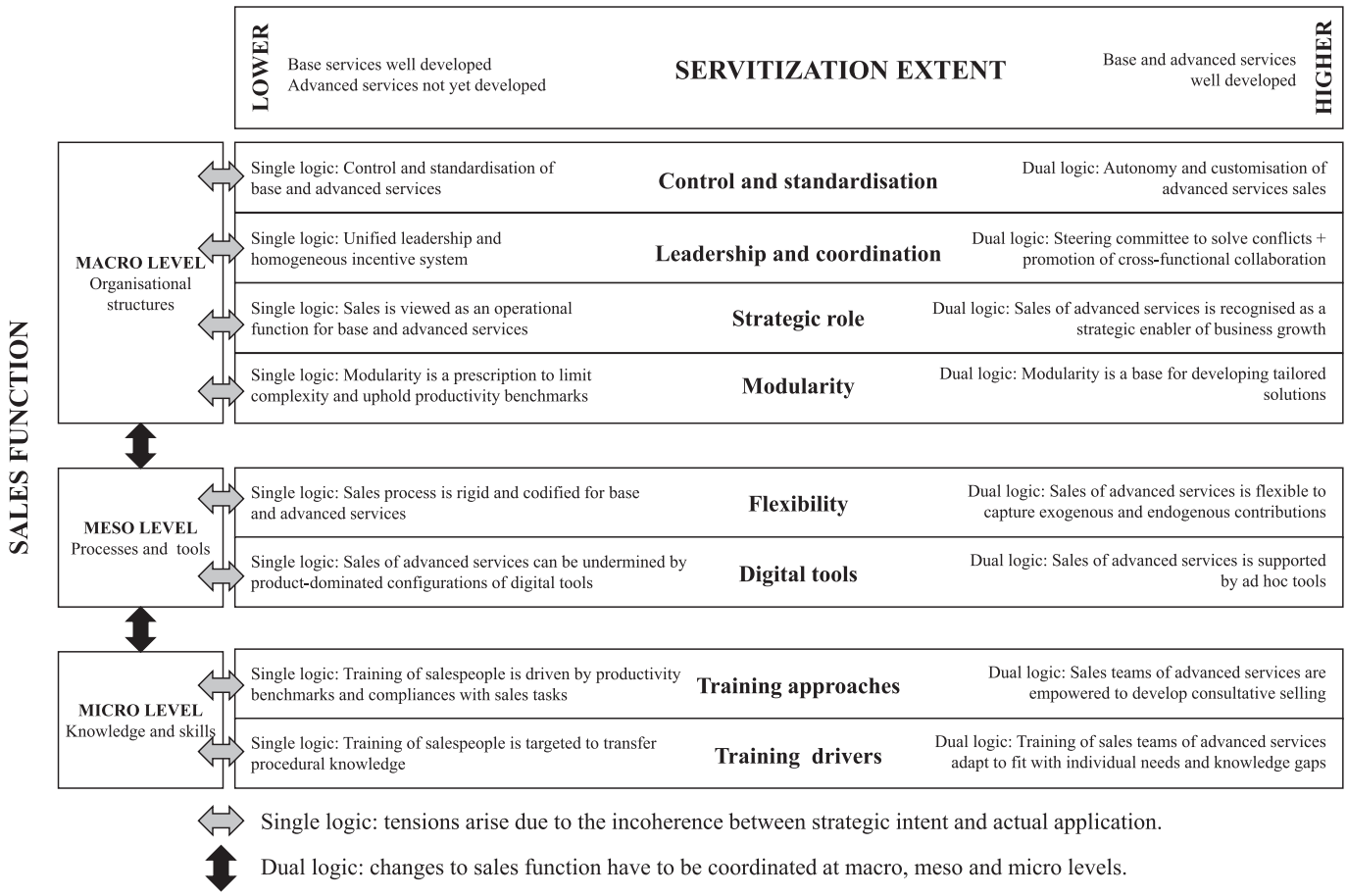

Fig. 1. Multilevel framework for sales function transformation in servitizing firms.

routines (Tuli et al., 2007; Witell et al., 2020), or sales capabilities (Terho et al., 2012; Ulaga & Loveland, 2014)—few have addressed their interplay. This research shows that transformation does not unfold linearly through isolated change initiatives but rather requires recursive alignment across organisational layers. Misalignments—such as standardised processes conflicting with consultative sales skills—emerge as structural tensions that constrain change. By embedding these tensions within a multilevel learning trajectory (Kowalkowski et al., 2025; Kozlowski & Klein, 2000), the framework clarifies how successful servitization depends on achieving coherence between strategic intent, operational systems, and frontline practices.

The second contribution of this study is to advance theorising on hybrid sales configurations in servitized manufacturing by explaining how the coexistence of base and advanced service logics is organisationally enacted within the sales function. Prior research has established that servitized manufacturers typically operate hybrid service portfolios in which base and advanced services coexist (Kowalkowski et al., 2015; Storbacka et al., 2013; Visnjic et al., 2016), and that this coexistence poses significant challenges for sales organisations (Kowalkowski et al., 2025; Salonen & Jaakkola, 2015; Ulaga & Reinartz, 2011). Existing studies have primarily addressed hybridity by either highlighting the strategic relevance of offering both service types or by examining organisational structures such as separation between sales units dedicated to base and advanced services (Favoretto et al., 2022; Kindström & Kowalkowski, 2014; Reinartz & Ulaga, 2008; Sharma, 2007). However, this body of work provides more limited theorisation of the organisational mechanisms through which the coexistence of base and advanced service logics is coordinated and sustained within sales organisations' activities (Kowalkowski et al., 2025). This study extends prior work by specifying hybridity as an outcome of coordinated organisational design choices across sales structures, processes, tools, and skills development. In doing so, it shifts the focus from whether base and advanced service logics should coexist to how sales organisations actively configure and govern this coexistence over time. By shifting attention from the presence of hybrid service portfolios to the organisational conditions under which base and advanced service logics are jointly enacted, the study complements and refines existing theorisation of sales organisation in servitized firms (Reinartz & Ulaga, 2008; Sharma, 2007) and contributes to servitization theory by clarifying the organisational foundations that enable hybrid sales systems to function effectively under conditions of service complexity.

The third contribution of this study is to advance a systemic view of sales function capability development in servitized manufacturing by explaining how capabilities supporting advanced services consolidate when organisational legitimacy and decision autonomy are aligned with dedicated digital tools across the customer journey. Prior research on servitized and solution selling has primarily examined sales capability development by focusing on the skills and competencies required to support advanced services, such as customer interaction and relationship management (Salonen et al., 2021; Sharma, 2007; Terho et al., 2012). While this literature has clarified the nature of the capabilities associated with advanced service selling, it has provided more limited insight into how such capabilities are organisationally supported and sustained within the sales function over time (Kowalkowski et al., 2017; Ulaga & Loveland, 2014). This study extends prior work by conceptualising sales function capability development as a systemic organisational phenomenon, rather than as the outcome of individual development initiatives alone. Specifically, it shows that capabilities supporting advanced services consolidate when sales teams are granted organisational legitimacy—understood as a recognised role in shaping

customer engagements and coordinating internal resources—and when this legitimacy is matched with appropriate decision autonomy and dedicated digital tools that support sales activities across base and advanced services (Kowalkowski et al., 2017; Rapaccini et al., 2023; Ritter & Pedersen, 2020). By linking capability development to the alignment of organisational legitimacy, autonomy, and digital enablement, this study contributes to servitization research by clarifying the organisational foundations through which sales functions can support advanced services within hybrid service portfolios.

In sum, this study offers an integrated perspective that advances current theories of sales transformation in servitizing firms. By framing the process as one of multilevel alignment, it explains how transformation unfolds across interrelated domains and provides a theoretical foundation for future research to investigate the conditions under which hybrid sales approaches succeed or fail. The findings extend the scope of servitization research and offer new conceptual tools for understanding how firms reconfigure their sales capabilities in response to growing service complexity and hybrid market demands.

7.2. Managerial implications

This research offers relevant insights for managers engaged in servitization, who have to reshape their sales function and teams. First, the importance of adopting a dual structure of the sales function emerges as the most effective in distinguishing roles, processes and tools between the sale of base and advanced services. Conversely, homogeneous organisational configurations appear inadequate to support transactional and relational sales approaches at the same time. Managers are therefore encouraged to recognise and promote this structural ambidexterity, clearly defining mandates, objectives and metrics for different sales teams. Secondly, the research highlights the need to reskill salespersons selling advanced services with a focus on hybrid skills, which integrate technical, business, and communication skills. Training should evolve beyond standardised, compliance-driven programs toward more personalised and adaptive approaches. It is also essential to build organisational legitimacy paths for vendors of advanced services, equipping them with decision-making tools, access to data, and recognition in cross-functional processes. A third implication concerns the use of the modularity of the offer. Firms should revisit the use of their modular service offerings to support customisation by sales teams, thus avoiding the use of modularity as a tool that limits the potential of salespeople. Finally, digitalisation must be consistent with the level of complexity of the sales process. In advanced services, digital tools can enable co-creative and collaborative approaches, but only if accompanied by the introduction of customer-centric culture and well-defined customer-supplier people-based collaboration routines. In base services, it is necessary to invest in digital tools to enable productivity and compliance with sales policies, bridging the gap that exists today between technological potential and operational practices.

7.3. Limitations and directions for future research

Several limitations should be acknowledged. For each limitation, we outline corresponding directions for future research that could extend and refine the insights developed here. First, the analysis is based on two polar case studies of large manufacturing firms, selected through theoretical sampling to represent different stages in the servitization journey. While this approach allows for in-depth comparison across contrasting configurations, the limited number of cases inevitably constrains the generalisability of the findings. Future research could include a broader set of firms across different industrial sectors or organisational settings to validate and extend the insights developed in this study. Second, while our findings point to the importance of internal legitimisation mechanisms in enabling a consulting-oriented sales approach, our analysis primarily focuses on the sales function. This limits the possibility of exploring how other organisational actors or functions contribute to or resist the institutionalisation of this framework. Future studies could investigate more systematically how internal dynamics—such as leadership, strategic alignment, and performance management systems—shape the adoption and stabilisation of consulting approaches across different parts of the organisation. Third, the integration between sales and service innovation is identified as a key challenge in both cases. However, our cross-sectional design does not allow us to trace how this integration unfolds over time. Longitudinal research could provide valuable insights into the evolution of sales roles and capabilities, particularly in the early phases of ideation and design of advanced services. Finally, the impact of artificial intelligence and predictive tools on the sales function was not the main focus of our study, although both cases revealed an increasing organisational interest in this area. Further research could explore how these technologies support—or transform—human involvement in knowledge-intensive sales processes, especially in the context of advanced service offerings.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this manuscript, the authors used generative artificial intelligence tools (ChatGPT 5.2) solely to improve the readability and clarity of the text. Following the use of these tools, the authors critically reviewed and revised the content as necessary and assume full responsibility for the accuracy, integrity, and originality of the published work.

CRediT authorship contribution statement

Maria Spadafora: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mario Rapaccini:** Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.indmarman.2026.03.012.

Data availability

Data will be made available on request.

References

Allison, L., & Flaherty, K. E. (2020). Investigating firm level drivers of salesperson brand identification. *Journal of Business Research, 121*, 154–169. https://doi.org/10.1016/j.jbusres.2020.08.030

Ardolino, M., Rapaccini, M., Saccani, N., Gaiardelli, P., Crespi, G., & Ruggeri, C. (2018). The role of digital technologies for the service transformation of industrial companies. *International Journal of Production Research, 56*(6), 2116–2132. https://doi.org/10.1080/00207543.2017.1324224

Arioli, V., Pezzotta, G., Romero, D., Adrodegari, F., Sala, R., Rapaccini, M., Saccani, N., Marjanovic, U., Rakic, S., West, S., Stoll, O., Wiesner, S., Bertoni, M., Lopez Odriozola, U., Pirola, F., & Gaiardelli, P. (2025). Digital servitization business typologies in the manufacturing sector. *International Journal of Industrial Engineering and Management, 16*(1), 1–23. https://doi.org/10.24867/IJIEM-378

Arnett, D. B., & Badrinarayanan, V. (2005). Enhancing customer-needs–driven CRM strategies: Core selling teams, knowledge management competence, and relationship

marketing competence. *Journal of Personal Selling & Sales Management, 25*(4), 329–343. https://doi.org/10.1080/08853134.2005.10749068

Baines, T., Bigdeli, A. Z., & Kapoor, K. (2024). *Servitization strategy: Delivering customer-centric outcomes through business model innovation*. Palgrave Macmillan.

Baines, T., Bigdeli, A. Z., Sousa, R., & Schroeder, A. (2020). Framing the servitization transformation process: A model to understand and facilitate the servitization journey. *International Journal of Production Economics, 221*, Article 107463. https://doi.org/10.1016/j.ijpe.2019.07.036

Bigdeli Z., A., Baines, T., Bustinza, O. F., & Guang Shi, V. (2017). Organisational change towards servitization: A theoretical framework. Competitiveness Review, 27(1), 12–39. doi:https://doi.org/10.1108/CR-03-2015-0015.

Blocker, C. P., Cannon, J. P., Panagopoulos, N. G., & Sager, J. K. (2012). The role of the sales force in value creation and appropriation: New directions for research. *Journal of Personal Selling & Sales Management, 32*(1), 15–27. https://doi.org/10.2753/PSS0885-3134320103

Brady, T., Davies, A., & Gann, D. M. (2005). Creating value by delivering integrated solutions. *International Journal of Project Management, 23*(5), 360–365. https://doi.org/10.1016/j.ijproman.2005.01.001

Brax, S. A., Bask, A., Hsuan, J., & Voss, C. (2017). Service modularity and architecture–an overview and research agenda. *International Journal of Operations & Production Management, 37*(6), 686–702. https://doi.org/10.1108/IJOPM-03-2017-0191

Bryman, A. (2006). Integrating quantitative and qualitative research: How is it done? *Qualitative Research, 6*(1), 97–113. https://doi.org/10.1177/1468794106058877

Capanni, G., Rapaccini, M., Momeni, B., Knapp, U. T., & Poeppelbuss, J. (2025). A contingency perspective on global service development in manufacturing firms. *Industrial Marketing Management, 127*, 100–118. https://doi.org/10.1016/j.indmarman.2025.04.004

Cenamor, J., Sjödin, D. R., & Parida, V. (2017). Adopting a platform approach in servitization: Leveraging the value of digitalization. *International Journal of Production Economics, 192*, 54–65. https://doi.org/10.1016/j.ijpe.2016.12.033

Coreynen, W., Matthyssens, P., & Van Bockhaven, W. (2017). Boosting servitization through digitization: Pathways and dynamic resource configurations for manufacturers. *Industrial Marketing Management, 60*, 42–53. https://doi.org/10.1016/j.indmarman.2016.04.012

Cron, W. L., & Decarlo, T. E. (2010). *Sales management: Concepts and cases* (10th ed.). Hoboken, NJ: John Wiley & Sons.

Eggert, A., Ulaga, W., & Gehring, A. (2020). Managing customer success in business markets: Conceptual foundation and practical application. *SMR-Journal of Service Management Research, 4*(2–3), 121–132.

Eisenhardt, K. M. (1989). Building theories from case study research. *Academy of Management Review, 14*(4), 532–550. https://doi.org/10.5465/amr.1989.4308385

Eisenhardt, K. M., & Graebner, M. E. (2007). Theory building from cases: Opportunities and challenges. *Academy of Management Journal, 50*(1), 25–32. https://doi.org/10.5465/amj.2007.24160888

Favoretto, C., Mendes, G. H., Oliveira, M. G., Cauchick-Miguel, P. A., & Coreynen, W. (2022). From servitization to digital servitization: How digitalization transforms companies' transition towards services. *Industrial Marketing Management, 102*, 104–121. https://doi.org/10.1016/j.indmarman.2022.01.003

Fischer, H., Seidenstricker, S., & Poeppelbuss, J. (2023). The triggers and consequences of digital sales: A systematic literature review. *Journal of Personal Selling & Sales Management, 43*(1), 5–23. https://doi.org/10.1080/08853134.2022.2102029

Gebauer, H., & Kowalkowski, C. (2012). Customer-focused and service-focused orientation in organisational structures. *Journal of Business & Industrial Marketing, 27*(7), 527–537. https://doi.org/10.1108/08858621211257293

Gebauer, H., Saul, C. J., Haldimann, M., & Gustafsson, A. (2017). Organizational capabilities for pay-per-use services in product-oriented companies. *International Journal of Production Economics, 192*, 157–168. https://doi.org/10.1016/j.ijpe.2016.12.007

Gremyr, I., Löfberg, N., & Witell, L. (2010). Service innovations in manufacturing firms. *Managing Service Quality: An International Journal, 20*(2), 161–175. https://doi.org/10.1108/09604521011027589

Grönroos, C., & Voima, P. (2013). Critical service logic: Making sense of value creation and co-creation. *Journal of the Academy of Marketing Science, 41*, 133–150. https://doi.org/10.1007/s11747-012-0308-3

Kaski, T., Niemi, J., & Pullins, E. (2018). Rapport building in authentic B2B sales interaction. *Industrial Marketing Management, 69*, 235–252. https://doi.org/10.1016/j.indmarman.2017.08.019

Khanra, S., Dhir, A., Parida, V., & Kohtamäki, M. (2021). Servitization research: A review and bibliometric analysis of past achievements and future promises. *Journal of Business Research, 131*, 151–166. https://doi.org/10.1016/j.jbusres.2021.03.056

Kindström, D., & Kowalkowski, C. (2009). Development of industrial service offerings: A process framework. *Journal of Service Management, 20*(2), 156–172. https://doi.org/10.1108/09564230910952753

Kindström, D., & Kowalkowski, C. (2014). Service innovation in product-centric firms: A multidimensional business model perspective. *Journal of Business & Industrial Marketing, 29*(2), 96–111. https://doi.org/10.1108/JBIM-08-2013-0165

Kindström, D., Kowalkowski, C., & Alejandro, T. B. (2015). Adding services to product-based portfolios: An exploration of the implications for the sales function. *Journal of Service Management, 26*(3), 372–393. https://doi.org/10.1108/JOSM-02-2014-0042

Kindström, D., Kowalkowski, C., & Sandberg, E. (2013). Enabling service innovation: A dynamic capabilities approach. *Journal of Business Research, 66*(8), 1063–1073. https://doi.org/10.1016/j.jbusres.2012.03.003

Kohli, A. K., & Jaworski, B. J. (1990). Market orientation: The construct, research propositions, and managerial implications. *Journal of Marketing, 54*(2), 1–18. https://doi.org/10.1177/002224299005400201

Kohtamäki, M., Brekke, T., Naeem, R., Sjödin, D., & Parida, V. (2025). Managerial practices shaping the transformation towards artificial intelligence-enabled product–service systems. *International Journal of Production Economics*. , Article 109658. https://doi.org/10.1016/j.ijpe.2025.109658

Kohtamäki, M., Parida, V., Patel, P. C., & Gebauer, H. (2020). The relationship between digitalization and servitization: The role of servitization in capturing the financial potential of digitalization. *Technological Forecasting and Social Change, 151*, Article 119804. https://doi.org/10.1016/j.techfore.2019.119804

Koponen, J., Julkunen, S., & Asai, A. (2019). Sales communication competence in international B2B solution selling. *Industrial Marketing Management, 82*, 238–252. https://doi.org/10.1016/j.indmarman.2019.01.009

Kowalkowski, C., Gebauer, H., Kamp, B., & Parry, G. (2017). Servitization and deservitization: Overview, concepts, and definitions. *Industrial Marketing Management, 60*, 4–10. https://doi.org/10.1016/j.indmarman.2016.12.007

Kowalkowski, C., Kramer, V., Eravci, S., Salonen, A., & Ulaga, W. (2025). Selling and sales management for successful servitization: A systematic review and research agenda. *Journal of Personal Selling & Sales Management*, 1–27. https://doi.org/10.1080/08853134.2025.2502168

Kowalkowski, C., & Ulaga, W. (2017). *Service strategy in action: A practical guide for growing your B2B service and solution business*. Service strategy press.

Kowalkowski, C., Windahl, C., Kindström, D., & Gebauer, H. (2015). What service transition? Rethinking established assumptions about manufacturers' service-led growth strategies. *Industrial Marketing Management, 45*, 59–69. https://doi.org/10.1016/j.indmarman.2015.02.016

Kozlowski, S. W. J., & Klein, K. J. (2000). A multilevel approach to theory and research in organizations: Contextual, temporal, and emergent processes. In K. J. Klein, & S. W. J. Kozlowski (Eds.), *Multilevel theory, research and methods in organizations: Foundations, extensions, and new directions* (pp. 3–90). San Francisco, CA: Jossey-Bass.

Lenka, S., Parida, V., & Wincent, J. (2017). Digitalization capabilities as enablers of value co-creation in servitizing firms. *Psychology & Marketing, 34*(1), 92–100. https://doi.org/10.1002/mar.20975

Mastrogiacomo, L., Barravecchia, F., & Franceschini, F. (2019). A worldwide survey on manufacturing servitization. *The International Journal of Advanced Manufacturing Technology, 103*(9), 3927–3942. https://doi.org/10.1007/s00170-019-03740-z

Meredith, J. (1998). Building operations management theory through case and field research. *Journal of Operations Management, 16*, 441–454. https://doi.org/10.1016/S0272-6963(98)00023-0

Oliva, R., & Kallenberg, R. (2003). Managing the transition from products to services. *International Journal of Service Industry Management, 14*(2), 160–172. https://doi.org/10.1108/09564230310474138

Paiola, M., Grandinetti, R., Kowalkowski, C., & Rapaccini, M. (2024). Digital servitization strategies and business model innovation: The role of knowledge-intensive business services. *Journal of Engineering and Technology Management, 74*, Article 101846. https://doi.org/10.1016/j.jengtecman.2024.101846

Panagopoulos, N. G., Rapp, A. A., & Ogilvie, J. L. (2017). Salesperson solution involvement and sales performance: The contingent role of supplier firm and customer–supplier relationship characteristics. *Journal of Marketing, 81*(4), 144–164. https://doi.org/10.1509/jm.15.0342

Parida, V., & Jovanovic, M. (2022). Servitization in global markets: Role alignment in global service networks for advanced service provision. *R&D Management, 52*(3), 577–592. https://doi.org/10.1111/radm.12505

Paschou, T., Rapaccini, M., Adrodegari, F., & Saccani, N. (2020). Digital servitization in manufacturing: A systematic literature review and research agenda. *Industrial Marketing Management, 89*, 278–292. https://doi.org/10.1016/j.indmarman.2020.02.012

Payne, A. F., Storbacka, K., & Frow, P. (2008). Managing the co-creation of value. *Journal of the Academy of Marketing Science, 36*, 83–96. https://doi.org/10.1007/s11747-007-0070-0

Piercy, N. F. (2010). Evolution of strategic sales organisations in business-to-business marketing. *Journal of Business & Industrial Marketing, 25*(5), 349–359. https://doi.org/10.1108/08858621011058115

Piercy, N. F., & Lane, N. (2005). Strategic imperatives for transformation in the conventional sales organisation. *Journal of Change Management, 5*(3), 249–266. https://doi.org/10.1080/14697010500175094

Powers, T. L., Sheng, S., & Li, J. J. (2016). Provider and relational determinants of customer solution performance. *Industrial Marketing Management, 56*, 14–23. https://doi.org/10.1016/j.indmarman.2016.03.006

Rabetino, R., Harmsen, W., Kohtamäki, M., & Sihvonen, J. (2018). Structuring servitization-related research. *International Journal of Operations & Production Management, 38*(2), 350–371. https://doi.org/10.1108/IJOPM-03-2017-0175

Raddats, C., & Burton, J. (2011). Strategy and structure configurations for services within product-centric businesses. *Journal of Service Management, 22*(4), 522–539. https://doi.org/10.1108/09564231111155105

Raddats, C., Kowalkowski, C., Benedettini, O., Burton, J., & Gebauer, H. (2019). Servitization: A contemporary thematic review of four major research streams. *Industrial Marketing Management, 83*, 207–223. https://doi.org/10.1016/j.indmarman.2019.03.015

Raja, J. Z., Bourne, D., Goffin, K., Çakkol, M., & Martinez, V. (2013). Achieving customer satisfaction through integrated products and services: An exploratory study. *Journal of Product Innovation Management, 30*(6), 1128–1144. https://doi.org/10.1111/jpim.12050

Raja, J. Z., Frandsen, T., Kowalkowski, C., & Jarmatz, M. (2020). Learning to discover value: Value-based pricing and selling capabilities for services and solutions. *Journal of Business Research, 114*, 142–159. https://doi.org/10.1016/j.jbusres.2020.03.026

Rapaccini, M. (2015). Pricing strategies of service offerings in manufacturing companies: A literature review and empirical investigation. *Production Planning & Control, 26* (14–15), 1247–1263. https://doi.org/10.1080/09537287.2015.1033495

Rapaccini, M., Paiola, M., Cinquini, L., & Giannetti, R. (2023). Digital servitization journey in small-and medium-sized enterprises: The contribution of knowledge-intensive business firms. *Journal of Business & Industrial Marketing, 38*(6), 1362–1375. https://doi.org/10.1108/JBIM-01-2022-0008

Rapaccini, M., Saccani, N., Kowalkowski, C., Paiola, M., & Adrodegari, F. (2020). Navigating disruptive crises through service-led growth: The impact of COVID-19 on Italian manufacturing firms. *Industrial Marketing Management, 88*, 225–237. https://doi.org/10.1016/j.indmarman.2020.05.017

Rapaccini, M., & Visintin, F. (2015). Devising hybrid solutions: An exploratory framework. *Production Planning & Control, 26*(8), 654–672. https://doi.org/10.1080/09537287.2014.961106

Reinartz, W., & Ulaga, W. (2008). How to sell services more profitably. *Harvard Business Review, 86*(5), 90–96.

Ritter, T., & Pedersen, C. L. (2020). Digitization capability and the digitalization of business models in business-to-business firms: Past, present, and future. *Industrial Marketing Management, 86*, 180–190. https://doi.org/10.1016/j.indmarman.2019.11.019

Robson, C. (2002). *Real world research*. Hoboken, NJ, USA: John Wiley and Sons.

Salonen, A., & Jaakkola, E. (2015). Firm boundary decisions in solution business: Examining internal vs. external resource integration. *Industrial Marketing Management, 51*, 171–183. https://doi.org/10.1016/j.indmarman.2015.05.002

Salonen, A., Terho, H., Böhm, E., Virtanen, A., & Rajala, R. (2021). Engaging a product-focused sales force in solution selling: Interplay of individual-and organizational-level conditions. *Journal of the Academy of Marketing Science, 49*(1), 139–163. https://doi.org/10.1007/s11747-020-00729-z

Sarasvuo, S., Liljander, V., & Haahtela, K. (2023). Buyer perceptions of corporate brand extension attractiveness and fit in B2B services. *Industrial Marketing Management, 115*, 69–85. https://doi.org/10.1016/j.indmarman.2023.09.006

Selnes, F. (1998). Antecedents and consequences of trust and satisfaction in buyer-seller relationships. *European Journal of Marketing, 32*(3/4), 305–322. https://doi.org/10.1108/03090569810204580

Shams, R., Chatterjee, S., & Chaudhuri, R. (2024). Developing brand identity and sales strategy in the digital era: Moderating role of consumer belief in brand. *Journal of Business Research, 179*, Article 114689. https://doi.org/10.1016/j.jbusres.2024.114689

Sharma, A. (2007). The shift in sales organisations in business-to-business services markets. *Journal of Services Marketing, 21*(5), 326–333. https://doi.org/10.1108/08876040710773633

Sheth, J. N., & Sharma, A. (2008). The impact of the product to service shift in industrial markets and the evolution of the sales organisation. *Industrial Marketing Management, 37*(3), 260–269. https://doi.org/10.1016/j.indmarman.2007.07.010

Sjödin, D., Parida, V., & Kohtamäki, M. (2019). Relational governance strategies for advanced service provision: Multiple paths to superior financial performance in servitization. *Journal of Business Research, 101*, 906–915. https://doi.org/10.1016/j.jbusres.2019.02.042

Sklyar, A., Kowalkowski, C., Tronvoll, B., & Sörhammar, D. (2019). Organizing for digital servitization: A service ecosystem perspective. *Journal of Business Research, 104*, 450–460. https://doi.org/10.1016/j.jbusres.2019.02.012

Sklyar, A., Sörhammar, D., Kowalkowski, C., & Tronvoll, B. (2025). Digital service orientation: Unlocking servitization in service operations and service sales. *International Journal of Operations & Production Management, 45*(13), 117–139. https://doi.org/10.1108/IJOPM-08-2024-0710

Spadafora, M., & Rapaccini, M. (2024). Bridging the gap between servitization and social innovation. *Journal of Cleaner Production.* , Article 142178. https://doi.org/10.1016/j.jclepro.2024.142178

Spadafora, M., Rapaccini, M., & West, S. (2024). Defining the smartness of data-driven PSS in decision-making. In *Data-driven decision making for product service systems* (pp. 225–243). Cham: Springer International Publishing. https://doi.org/10.1007/978-3-031-60082-1_10.

Spring, M., & Araujo, L. (2013). Beyond the service factory: Service innovation in manufacturing supply networks. *Industrial Marketing Management, 42*(1), 59–70. https://doi.org/10.1016/j.indmarman.2012.11.006

Storbacka, K. (2011). A solution business model: Capabilities and management practices for integrated solutions. *Industrial Marketing Management, 40*(5), 699–711. https://doi.org/10.1016/j.indmarman.2011.05.003

Storbacka, K., Ryals, L., Davies, I. A., & Nenonen, S. (2009). The changing role of sales: Viewing sales as a strategic, cross-functional process. *European Journal of Marketing, 43*(7/8), 890–906. https://doi.org/10.1108/03090560910961443

Storbacka, K., Windahl, C., Nenonen, S., & Salonen, A. (2013). Solution business models: Transformation along four continua. *Industrial Marketing Management, 42*(5), 705–716. https://doi.org/10.1016/j.indmarman.2013.05.008

Teece, D. J., Pisano, G., & Shuen, A. (1997). Dynamic capabilities and strategic management. *Strategic Management Journal, 18*(7), 509–533. https://doi.org/10.1002/(SICI)1097-0266(199708)18:7%3C509::AID-SMJ882%3E3.0.CO;2-Z

Terho, H., Haas, A., Eggert, A., & Ulaga, W. (2012). 'It's almost like taking the sales out of selling'—Towards a conceptualization of value-based selling in business markets. *Industrial Marketing Management, 41*(1), 174–185. https://doi.org/10.1016/j.indmarman.2011.11.011

Tippmann, E., Monaghan, S., & Reuber, R. A. (2023). Navigating the paradox of global scaling. *Global Strategy Journal, 13*(4), 735–773. https://doi.org/10.1002/gsj.1435

Tronvoll, B., Sklyar, A., Sörhammar, D., & Kowalkowski, C. (2020). Transformational shifts through digital servitization. *Industrial Marketing Management, 89*, 293–305. https://doi.org/10.1016/j.indmarman.2020.02.005

Tuli, K. R., Kohli, A. K., & Bharadwaj, S. G. (2007). Rethinking customer solutions: From product bundles to relational processes. *Journal of Marketing, 71*(3), 1–17. https://doi.org/10.1509/jmkg.71.3.001

Ulaga, W., & Eggert, A. (2006). Value-based differentiation in business relationships: Gaining and sustaining key supplier status. *Journal of Marketing, 70*(1), 119–136. https://doi.org/10.1509/jmkg.70.1.119.qx

Ulaga, W., & Kohli, A. K. (2018). The role of a solutions salesperson: Reducing uncertainty and fostering adaptiveness. *Industrial Marketing Management, 69*, 161–168. https://doi.org/10.1016/j.indmarman.2017.11.008

Ulaga, W., & Loveland, J. M. (2014). Transitioning from product to service-led growth in manufacturing firms: Emergent challenges in selecting and managing the industrial sales force. *Industrial Marketing Management, 43*(1), 113–125. https://doi.org/10.1016/j.indmarman.2013.08.006

Ulaga, W., & Reinartz, W. J. (2011). Hybrid offerings: How manufacturing firms combine goods and services successfully. *Journal of Marketing, 75*(6), 5–23. https://doi.org/10.1509/jm.09.0395

Verbeke, W., Dietz, B., & Verwaal, E. (2011). Drivers of sales performance: A contemporary meta-analysis. Have salespeople become knowledge brokers? *Journal of the Academy of Marketing Science, 39*, 407–428. https://doi.org/10.1007/s11747-010-0211-8

Visnjic, I., Wiengarten, F., & Neely, A. (2016). Only the brave: Product innovation, service business model innovation, and their impact on performance. *Journal of Product Innovation Management, 33*(1), 36–52. https://doi.org/10.1111/jpim.12254

Windahl, C., & Lakemond, N. (2010). Integrated solutions from a service-centered perspective: Applicability and limitations in the capital goods industry. *Industrial Marketing Management, 39*(8), 1278–1290. https://doi.org/10.1016/j.indmarman.2010.03.001

Windler, K., Jüttner, U., Michel, S., Maklan, S., & Macdonald, E. K. (2017). Identifying the right solution customers: A managerial methodology. *Industrial Marketing Management, 60*, 173–186. https://doi.org/10.1016/j.indmarman.2016.03.004

Witell, L., Kowalkowski, C., Perks, H., Raddats, C., Schwabe, M., Benedettini, O., & Burton, J. (2020). Characterizing customer experience management in business markets. *Journal of Business Research, 116*, 420–430. https://doi.org/10.1016/j.jbusres.2019.08.050

Yin, R. K. (2009). *Case study research: Design and methods*. Thousand Oaks, CA: Sage.

Zhang, W., & Banerji, S. (2017). Challenges of servitization: A systematic literature review. *Industrial Marketing Management, 65*, 217–227. https://doi.org/10.1016/j.indmarman.2017.06.003